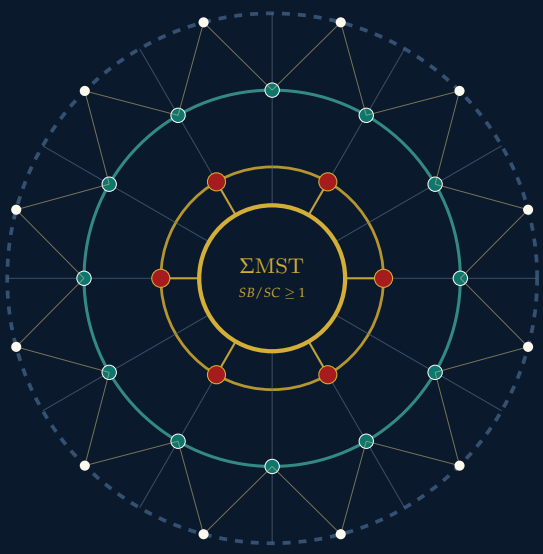
Post-Capitalist Labs Imprint · Working Draft Monograph

DISTRIBUTED PARTICIPATORY ECONOMICS PROTOCOL

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons



DESIGNED FOR THREE INTERDISCIPLINARY AUDIENCES:

- Advocates & Cooperative Practitioners: Council democracy, balanced job complexes, and historical lessons from Catalonia (1936), the Paris Commune (1871), and Classical Athens.
- Economists & Mechanism Designers: Formal welfare theorems, Sandström's double-entry accounting, pollution demand revelation (CAPs), and non-convex ADMM convergence.
- Systems Programmers & Cryptographers: Local-first CRDTs, BLAKE3 Sparse Merkle-Sum Trees, deterministic WASM IFB engines, and NIST FIPS 203/204/205 post-quantum security.

POST-CAPITALIST LABS IMPRINT · https://github.com/Post-Capitalist-Labs Draft Edition · Nov 2026

Leveraging technology to amplify social movements · Decentralised consensus, mechanism design, and democratic economic coordination

Post-Capitalist Labs Imprint · Open-Source Research Monograph Series

Distributed Participatory Economics Protocol (DPEP)

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons

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About Post-Capitalist Labs (PCL)

At Post-Capitalist Labs, we harness the transformative power of technology to bolster social movements and build directly democratic alternatives to capitalism. Our work spans decentralized consensus, applied cryptography, operations research, and democratic mechanism design.

Our Vision: We envisage a world where solidarity, care, classlessness, diversity, self-reflection, and direct democracy are not just ideals, but the foundations of everyday life. Our goal is to make this vision a reality through the thoughtful application of open-source technology, where people remain in control.

Our Approach:

- Learning from History: Analyzing past and present democratic movements and self-managed collectives to guide institutional design.
- Emancipatory Modeling: Developing rigorous mathematical and computational models for equitable, non-market resource coordination.
- Open-Source Engineering: Turning theoretical models into verifiable, accessible, community-driven proofs of concept and protocols.

Organization Profile: https://github.com/Post-Capitalist-Labs
Repository & Maintainer Profile: https://github.com/titancassini/distributed-participatory-economics-protocol

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Interdisciplinary Navigation Guide

How to Read This Book Across Disciplines

This monograph bridges libertarian-socialist institutional theory, formal welfare and planning economics, and modern distributed systems cryptography. Depending on your primary background, we recommend the following reading paths:

FOR ADVOCATES, COOPERATORS & ORGANIZERS — Institutional Self-Management Without Hierarchy

Start with Chapter 1 (Executive Summary) and Chapter 2 (Historical & Institutional Inspirations) to see how lessons from the 1936–1939 Spanish Revolution, the 1871 Paris Commune, Classical Athens, Occupy Wall Street, Mondragón, Porto Alegre Participatory Budgeting, Rojava, and Elinor Ostrom's commons principles are translated into concrete protocol safeguards against bureaucratic capture, meeting fatigue, and inter-regional inequality. Then read Chapter 3 for the complete traceability matrix linking Michael Albert and Robin Hahnel's 1991 and 2021 books to our design.

FOR ECONOMISTS & OPERATIONS RESEARCHERS — Formal Models, Accounting & Non-Convex Convergence

Focus on Chapter 4 (Formal Economic Model), which unites Albert and Hahnel's Formal Model of a Participatory Economy (FMPE) with Anders Sandström's double-entry Society Account and Robin Hahnel's Pollution Demand-Revealing Mechanism (PDRM). See Section 4.4 for our source-code audit of pequod-cljs and Algorithm 3 (Safeguarded Primal-Dual / ADMM Proximal Regularization), alongside Chapter 8 and Chapter 10 for our synthesis of John O'Neill's (2024) ecological incommensurability critique and non-convex convergence experiments.

FOR SYSTEMS PROGRAMMERS & CRYPTOGRAPHERS — Local-First P2P, Merkle-Sum Trees & Post-Quantum Security

Jump directly to Chapter 4 (Section 4.5 State Machine & Protobuf Schemas), Chapter 5 (Four-Layer Architecture D & 6-Role Mobile Taxonomy), Chapter 6 (Weighted Stack Comparison), Chapter 7 (NIST FIPS 203/204/205 Post-Quantum Cryptography, Non-Plutocratic Sybil Resistance & Zero-Knowledge Budget Proofs), and Chapter 9 (Implementable Proof-of-Concept Specification & Multi-Scale Benchmark Programme).

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List of Abbreviations and Mathematical Notation

Because this monograph integrates participatory economics, labor and cooperative history, operations research, distributed systems, and post-quantum cryptography, this reference table defines every abbreviation and mathematical symbol used across Chapters 1–10.

A. Participatory Economics, Accounting, and Bibliographic Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
BJC	Balanced Job Complex	Bundle of workplace tasks balanced for empowerment and desirability within and across workplaces (PEPE Ch. 2).
CAP	Community of Affected Parties	Ecological/geographic collective of residents impacted by a specific pollutant in an airshed or watershed (DEP Ch. 7).
CC (CC_h)	Consumers' Council	Democratic neighborhood consumer council or nested ward/city/regional consumer federation.
CFB	Consumption Facilitation Board	Informational board assisting consumer councils and federations in evaluating collective consumption projects (LF Ch. 6).
CLUEA	Consejo Levantino Unificado de Exportación Agrícola	Unified Levantine Council for Agricultural Export organized by the CNT and UGT in Valencia (1936–1939).
CNT	Confederación Nacional del Trabajo	Anarcho-syndicalist National Confederation of Labor in the 1936–1939 Spanish Revolution.
CPR	Common-Pool Resource	Shared natural or human-made resource system governed by collective-choice rules (Ostrom 1990).
DEP	Democratic Economic Planning	Primary reference volume by Robin Hahnel et al. (Routledge, 2021).
DPEP	Distributed Participatory Economics Protocol	The open-source decentralized coordination protocol specified in this monograph.
EFB	Employment Facilitation Board	Facilitation board assisting workers transferring between councils and balancing cross-workplace BJCs (LF Ch. 6).
EJPE	Erasmus Journal for Philosophy and Economics	Academic journal hosting the Winter 2024 symposium on Hahnel's Democratic Economic Planning.
FMPE	Formal Model of a Participatory Economy	Mathematical formalization of participatory planning analyzed in PEPE Chapter 5 (pp. 73–106).
FTE	Full-Time Equivalent	Unit of engineering or research staffing effort across roadmap scenarios.
GA	General Assembly	Direct plenary deliberative assembly of a workplace, village collective, or horizontal movement.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
GDP	Gross Domestic Product	Aggregate social value of net final output (Y^{GDP}) cleared in the Society Account.
GHG	Greenhouse Gas	Atmospheric emissions (CO_2-equivalent) subject to hard biophysical ceilings ($E_z^{\max}$).
HFB	Household Facilitation Board	Facilitation board assisting individuals and families with housing and neighborhood membership (LF Ch. 6).
IF (IF_k)	Industry Federation	Federation of worker councils within industry k managing sector capacity, R&D, and SB/SC appeals.
IFB	Iteration Facilitation Board	Informational board/engine that aggregates supply and demand and updates indicative prices via public rules.
LF	Looking Forward	Primary reference volume by Michael Albert and Robin Hahnel (South End Press, 1991).
Parecon / PE	Participatory Economics	Post-capitalist economic model developed by Michael Albert and Robin Hahnel.
PB	Participatory Budgeting / Planning Bureau	Municipal democratic budgeting (Porto Alegre, 1989); also denotes the abstract Planning Bureau in FMPE Ch. 5.
PCL	Post-Capitalist Labs	Open-source research and engineering collective publishing this monograph (https://github.com/Post-Capitalist-Labs).
PDRM	Pollution Demand-Revealing Mechanism	Iterative procedure pricing local pollutant damages via CAP emission tolerances and credits (DEP Ch. 7).
PEP	Participatory Economy Project	Research collective maintaining participatoryeconomy.org and the 2023 User Story Mapping.
PEPE	The Political Economy of Participatory Economics	Primary technical reference volume by Michael Albert and Robin Hahnel (Princeton University Press, 1991).
PLAC	Parecon: Life After Capitalism	Comprehensive institutional and daily-life synthesis of Participatory Economics by Michael Albert (Verso Books, 2003).
PFB	Production Facilitation Board	Board assisting worker councils with production externalities, scale economies, and technology choice (LF Ch. 6).
PIPA	Practically Identical Production Activities	Equivalence classes $\{PIPA\}_i$ partitioning workplace tasks by qualitative empowerment effect (PEPE Ch. 2, p. 26).
SB / SC	Social Benefit / Social Cost	Ratio $SB_j(P)/SC_j(P)$ measuring the social value of a worker council's outputs relative to the opportunity cost of its inputs and endowments (≥ 1 constraint).
SKU	Stock Keeping Unit	Granular product variant (size, color, style) resolved at intra-year fulfillment from coarse planning categories.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
UFB	Updating Facilitation Board	Facilitation board mediating intra-year plan adjustments when demand shifts or supply shocks exceed slack (LF Ch. 6).
UGT	Unión General de Trabajadores	Socialist General Union of Workers that co-organized industrial and agrarian collectives in Spain (1936–1939).
WC (WC_j)	Workers' Council	Democratic workplace collective where each worker has one vote and holds a Balanced Job Complex.

B. Operations Research, AI, Distributed Systems, and Cryptography Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
ADMM	Alternating Direction Method of Multipliers	Augmented-Lagrangian primal-dual optimization method with proximal damping used in Algorithm 3.
AP / CP	Availability / Consistency under Partition	CAP theorem properties: local-first drafting is AP (available offline), while Epoch BFT price finality is CP.
ARIMA / ETS	AutoRegressive Integrated Moving Average / Error-Trend-Seasonality	Classical statistical time-series forecasting models used for baseline demand and SKU disaggregation.
BFT	Byzantine Fault Tolerance	Distributed consensus property guaranteeing safety even when up to $f < n/3$ validator nodes act maliciously.
BLAKE3	BLAKE3 Cryptographic Hash	High-speed 256-bit cryptographic hash function used for Sparse Merkle-Sum Trees and content addressing.
BSP	Bulk-Synchronous Parallel	Execution model alternating parallel local computation rounds with synchronized global aggregation checkpoints.
CID	Content Identifier	Self-certifying cryptographic hash address referencing an immutable qualitative List or plan blob.
CRDT	Conflict-Free Replicated Data Type	Data structure (Loro / Automerge) allowing offline concurrent edits in council workspaces that merge deterministically.
CRQC	Cryptographically Relevant Quantum Computer	Future fault-tolerant quantum computer capable of breaking classical elliptic-curve and RSA cryptography via Shor's algorithm.
DAO	Decentralized Autonomous Organization	Blockchain governance construct critiqued in Chapter 2 for conflating token voting with council democracy.
DTN	Delay-Tolerant Networking	Store-and-forward networking (including LoRa mesh and USB data-mules) for intermittently connected regions.
EEIO / LCA	Environmentally Extended Input-Output / Lifecycle Assessment	Matrix accounting methods estimating cradle-to-gate resource and emission footprints across supply chains.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
FIPS	Federal Information Processing Standards	NIST post-quantum cryptographic standards finalized in August 2024: FIPS 203, FIPS 204, and FIPS 205.
FRI / STARK	Fast Reed-Solomon Interactive Oracle Proof / Scalable Transparent ARgument of Knowledge	Post-quantum, hash-based zero-knowledge proof system (Plonky3) requiring no trusted setup.
IEEE-754	IEEE Floating-Point Standard	Hardware floating-point representation replaced in DPEP core rules by deterministic 64-bit fixed-point integers.
KKT	Karush-Kuhn-Tucker Conditions	First-order necessary and sufficient optimality conditions for constrained nonlinear/conic optimization.
LP / QP / MILP	Linear / Quadratic / Mixed-Integer Linear Programming	Mathematical optimization paradigms solved locally via Clarabel.rs and HiGHS.
MAPE	Mean Absolute Percentage Error	Accuracy metric used to audit and backtest statistical demand forecasts.
ML-DSA-65	Module-Lattice Digital Signature Algorithm	NIST FIPS 204 (Dilithium3) post-quantum digital signature scheme combined in hybrid mode with Ed25519.
ML-KEM-768	Module-Lattice Key-Encapsulation Mechanism	NIST FIPS 203 (Kyber-768) post-quantum key-establishment scheme combined in hybrid mode with X25519.
MLS	Messaging Layer Security	IETF RFC 9420 protocol providing end-to-end encrypted group messaging inside councils.
MPC	Secure Multiparty Computation	Cryptographic paradigm compared against lightweight pairwise additive masking in Chapters 6–7.
MST	Sparse Merkle-Sum Tree	Authenticated tree binding both BLAKE3 hashes and non-negative supply/demand vector sums at every federation node.
NASA-TLX	NASA Task Load Index	Standardized human-factors metric used to measure cognitive load and participation fatigue in PoC testing.
NMT / TTS	Neural Machine Translation / Text-to-Speech	Local accessibility models translating qualitative Lists across languages and assisting visually impaired users.
OR	Operations Research	Discipline of mathematical optimization, linear programming, and supply-chain clearing.
P2P	Peer-to-Peer	Decentralized networking architecture without central client-server gatekeepers.
PoC	Proof of Concept	The 1,000-participant, 40-council, 80-good end-to-end reference implementation specified in Chapter 9.
PQ-UCAN	Post-Quantum User Controlled Authorization Network	Chained, cryptographically signed capability certificates encoding council membership and revocable delegation.
PQC	Post-Quantum Cryptography	Cryptographic algorithms resistant to both classical and quantum (Shor/Grover) attacks.
PWA	Progressive Web Application	Offline-capable cross-platform web client running the Rust/WASM Role-LP engine on smartphones and desktops.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
QUIC	Quick UDP Internet Connections	Modern encrypted, multiplexed transport protocol (quinn / iroh) used for Layer 2 federation sync.
Role-LP / CC / SR / PCW / BFT / GW	DPEP Node Roles 1–6	Lightweight Participant (LP), Council Coordinator (CC), Storage Replica (SR), Planning Compute Worker (PCW), Epoch Validator (BFT), and Delay-Tolerant Gateway (GW).
SLH-DSA	Stateless Hash-Based Digital Signature Algorithm	NIST FIPS 205 (SPHINCS+) signature standard relying solely on hash functions, used for long-term root anchors.
TPS	Transactions Per Second	Throughput metric shown in Chapter 5 to be irrelevant to epoch-synchronized participatory planning.
TTL	Time-To-Live	Hard expiration timer on delegate capability certificates and emergency replanning playbooks.
UniFFI	Universal Foreign Function Interface	Mozilla binding generator exposing the Rust dpep-core library to native Kotlin (Android) and Swift (iOS).
VRF	Verifiable Random Function	Cryptographic sortition primitive generating publicly verifiable random selections for audit and oversight boards.
WASM	WebAssembly	Deterministic, sandboxed binary instruction format (Wasmtime) executing IFB price-update rules.
ZKP	Zero-Knowledge Proof	Cryptographic proof verifying that a household's private consumption bundle obeys its budget $SC \leq B$ without revealing items.

C. Summary of Core Mathematical Notation (Chapter 4)

Symbol	Mathematical Meaning
Ω	Master catalog of all $G = \lvert\Omega\rvert$ priced economic categories: $\mathcal{G}^{\text{priv}} \sqcup \mathcal{G}^{\text{pub}} \sqcup \mathcal{G}^{\text{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z}$.
$P^{(t)} \in \mathbb{R}_+^{\lvert\Omega\rvert}$	Vector of indicative social opportunity-cost prices at planning iteration t.
$\bar{a}_j, \underline{a}_j, r_j, e_j$	Worker council WC_j's proposed outputs ($\bar{a}_j$), intermediate inputs ($\underline{a}_j$), natural resources (r_j), and emissions (e_j).
$l_j^0, k_j^0, \Delta l_j, \Delta k_j$	Initial labor (l_j^0) and capital (k_j^0) endowments of WC_j, and proposed net additions/releases ($\Delta l_j, \Delta k_j$).
$r^*, l^*, k^*, E_z^{\max}$	Economy-wide available natural resources (r^*), labor supplies (l^*), capital stocks (k^*), and hard biophysical emission caps ($E_z^{\max}$).
$w_{\text{base}}, \epsilon_u, B_u(P)$	Base hourly remuneration rate (w_{base}), worker effort multiplier (ϵ_u), and citizen u's warranted annual consumption budget (B_u).
$S_i^{(t)}, D_i^{(t)}, SL_i^{(t)}$	Aggregate supply ($S_i^{(t)}$), aggregate demand ($D_i^{(t)}$), and planned slack buffer ($SL_i^{(t)} = \eta_i S_i^{(t)}$) for category i.
$\rho_i^{(t)}, \tau$	Normalized relative excess demand imbalance $\rho_i^{(t)} = \frac{2\lvert S_i^{(t)} - D_i^{(t)}\rvert}{S_i^{(t)} + D_i^{(t)}}$ and feasibility convergence threshold $\tau \in [0.01, 0.05]$.

CHAPTER 1

Executive Summary: Feasibility, Architecture, and Strategic Roadmap

> "No central agency composes the final plan. No market competition generates the final plan. A decentralized, social, iterative, communication process allows all actors to democratically formulate the final plan."
>
> — Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (1991), Ch. 6[1]

1.1 What Is Being Proposed

This monograph presents a rigorous, source-grounded feasibility study and technical architecture proposal for the Distributed Participatory Economics Protocol (DPEP): an open-source, local-first, cryptographically verifiable coordination protocol enabling democratic worker councils, neighborhood consumer councils, and nested federations to formulate, iterate, ratify, and execute comprehensive economic plans without capitalist markets or authoritarian central planning.

The intellectual foundation of this protocol is Participatory Economics (Parecon), originally developed by Michael Albert and Robin Hahnel in The Political Economy of Participatory Economics (1991) and Looking Forward (1991),[2] elaborated across daily-life institutions and critic rebuttals in Michael Albert's Parecon: Life After Capitalism (2003),[3] and mathematically extended in Robin Hahnel's Democratic Economic Planning (2021),[4] Anders Sandström's Anarchist Accounting (2020),[5] and Mitchell Szczepanczyk's computer simulation work (pequod-cljs,

[1] Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (Boston, MA: South End Press, 1991), Ch. 6 ("Participatory Allocation"), accessed via ZNetwork authorized online edition (https://znetwork.org/wp-content/uploads/zbooks/htdocs/ParEcon/lookingforward/lf6_1.htm).

[2] Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (Princeton, NJ: Princeton University Press, 1991), https://doi.org/10.2307/j.ctv131bvr2; Albert and Hahnel, Looking Forward (1991).

[3] Michael Albert, Parecon: Life After Capitalism (London & New York: Verso Books, 2003), full text archived at ZCommunications/ZNetwork (http://books.zcommunications.org/books/pareconv/).

[4] Robin Hahnel (with contributions by Peter Bohmer, Savvina Chowdhury, Allison Kerkhoff, Mitchell Szczepanczyk, and Michael Weisdorf), Democratic Economic Planning (Abingdon & New York: Routledge, 2021), https://doi.org/10.4324/9781003173700.

[5] Anders Sandström, Anarchist Accounting: Accounting Principles for a Democratic Economy (Abingdon & New York: Routledge, 2020), https://doi.org/10.4324/9781003036005; see also Sandström, "An Accounting System for a Libertarian Socialist Economy," Participatory Economy Project (Feb. 14, 2023), https://participatoryeconomy.org/an-accounting-system-for-a-libertarian-socialist-economy/.

2023).[6]

DPEP provides the computational substrate for four core institutional pillars:

1. Self-Managed Work and Consumption: Democratic worker councils (WC_j) and consumer councils (CC_h) in which participants exercise decision-making input proportional to the degree they are affected by an economic choice.[7]
2. Equitable Remuneration and Balanced Job Complexes (BJCs): Non-transferable consumption allowances calibrated to the duration, intensity, and onerousness (effort and sacrifice) of socially valued labor, supplemented by universal free public goods and council-approved special-need allowances, paired with job complexes balanced for empowerment and desirability within and across workplaces.[8]
3. Decentralised Iterative Planning: A multi-round coordination protocol in which councils submit production and consumption proposals, federated aggregators compute excess demands across seven economic categories (private goods, public goods, intermediate inputs, natural resources, labor categories, capital stocks, and pollutant emissions), and a deterministic Iteration Facilitation Board (IFB) engine updates indicative social opportunity-cost prices until a feasible, ecologically bounded plan is ratified.[9]
4. Intra-Year Operational Clearing and Scoped Replanning: Continuous tracking of deliveries and inventory drawdowns against the ratified plan, absorbing routine consumer preference shifts locally within neighborhood and city slack buffers while mediating larger supply shocks across affected industry federations via Updating Facilitation Boards (UFBs).[10]

1.2 Why It Matters: Breaking the "No Third Way" Dogma

For over a century, mainstream and socialist economists alike have been haunted by Alec Nove's famous assertion in The Economics of Feasible Socialism (1983):

> "In a complex industrial economy the interrelation between its parts can be based in principle either on freely chosen negotiated contracts [i.e., markets],

[6] Mitchell Szczepanczyk, "Pseudocode and Algorithms for Computer Simulations of Democratically Planned Economies," Journal of Information Economics 1, no. 3 (2023): 43–54, https://doi.org/10.58567/jie01030004; open-source repository at https://github.com/msszczep/pequod-cljs.

[7] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 8, 16; Ch. 2, p. 24; Hahnel, Democratic Economic Planning, Ch. 1, pp. 14–16.

[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 25–30; Ch. 3, pp. 44–55.

[9] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, pp. 57–72; Ch. 5, pp. 73–106; Hahnel, Democratic Economic Planning, Ch. 7, pp. 130–162.

[10] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, pp. 66–67; Looking Forward, Ch. 6 (lf6_1.htm); Hahnel, Democratic Economic Planning, Ch. 8, pp. 167–170.

> or on a system of binding instructions from planning offices [i.e., command planning]. There is no third way."[11]

Yet historical experiments in democratic self-management—most notably the industrial and agrarian collectives of the Spanish Revolution (1936–1939), the Paris Commune (1871), federated cooperatives like Mondragón, and horizontal movements like Occupy Wall Street (2011)—repeatedly encountered two structural barriers when attempting to coordinate beyond single workplaces or rural villages:

- The Accounting and Valuation Barrier: Without a unified social accounting and indicative-price system, wealthier collectives hoarded superior inherited machinery (what Gaston Leval and Frank Mintz termed "collective egoism" or neocapitalismo colectivo in 1936 Barcelona),[12] while inter-industry coordination of complex intermediate inputs broke down into cumbersome bilateral barter.
- The Deliberative Scaling Barrier: Without structured, asynchronous computational protocols, horizontal deliberation degenerated into exhausting multi-hour plenary meetings with $\mathcal{O}(N^2)$ communication friction or unaccountable informal cliques (Jo Freeman's "tyranny of structurelessness").[13]

Contemporary distributed computing, post-quantum cryptography, and mathematical optimization make it possible to overcome both barriers simultaneously—allowing millions of participants to coordinate complex industrial supply chains and ecological boundaries democratically in seconds of computation per round rather than months of bureaucratic paperwork.

1.3 Technical Feasibility: Achievable Guarantees vs. Aspirational Properties

A core mandate of our investigation is to distinguish demonstrably achievable engineering guarantees from aspirational or conditionally achievable properties requiring experimental validation.

[11] Alec Nove, The Economics of Feasible Socialism (London: George Allen & Unwin, 1983), pp. ix–x, 44; quoted directly at the opening of Chapter 4 in Albert and Hahnel, Political Economy of Participatory Economics, p. 57.

[12] Gaston Leval, Collectives in the Spanish Revolution, trans. Vernon Richards (London: Freedom Press, 1975); Frank Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, trans. Paul Sharkey (Oakland, CA: AK Press, 2013); Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), Ch. 6.

[13] Jo Freeman, "The Tyranny of Structurelessness," The Second Wave 2, no. 1 (1972) / Berkeley Journal of Sociology 17 (1972–1973): 151–164; David Graeber, The Democracy Project: A History, a Crisis, a Movement (New York: Spiegel & Grau, 2013).

[14] Leonid Hurwicz, "The Design of Mechanisms for Resource Allocation," American Economic Review 63, no. 2 (1973): 1–30.

Table 1.1: Demonstrably Achievable Guarantees vs. Conditionally Achievable Properties

Demonstrably Achievable Today (Proven by Math & Systems Engineering)	Aspirational / Conditionally Achievable (Requires Experimental Validation)
1. Sub-10-Second Global Aggregation & Price Updates: Even for $1,000,000$ participants ($40,000$ councils, $10,000$ product categories), aggregating sparse council proposals up a federated tree and updating prices takes < 10 s per round across modest community servers.	1. Convergence Speed Under Severe Industrial Non-Convexities: Existing Parecon simulations (pequod-cljs, Hahnel et al. 2021) proved 6–12 round convergence for smooth Cobb-Douglas agents, but our source audit revealed they did not test fixed setup costs, Leontief complementarities, or endogenous income feedback.
2. Verifiable Neutrality of the IFB: Compiling the IFB price-update rule to a deterministic fixed-point WebAssembly (WASM) module paired with a BLAKE3 Sparse Merkle-Sum Tree (MST) mathematically prevents IFB operators from skewing prices or censoring council proposals.	2. Low-Fatigue Human Participation at Scale: Keeping household consumption planning under 30–60 minutes per year and preventing council burnout requires human-in-the-loop UX testing of automated drafting assistants and milestone voting schedules.
3. Offline-First & Low-End Smartphone Participation: A lightweight client (Role-LP) using SQLite, delta-CRDTs, and Zstd-compressed price deltas runs on a 5-year-old Android phone with < 60 MB storage and < 150 KB bandwidth per round.	3. Harmonious Workplace Effort Rating: Software can enforce the cap that a workplace's average effort rating cannot exceed $100 \times (SB_j / SC_j)$, but preventing interpersonal friction in peer evaluation is a social/institutional challenge.
4. Post-Quantum & Privacy-Preserving Household Budgets: Hybrid Ed25519 + ML-DSA-65 (FIPS 204) signatures, X25519 + ML-KEM-768 (FIPS 203), and pairwise additive masking with local budget proofs (Plonky3 STARKs) let a council sum private demands without neighbors seeing personal/medical items.	4. Truthful Revelation of Tacit Workplace Potential: While SB/SC benchmarking and public technology blueprints deter gross slacking, no mechanism can mathematically force a workplace to reveal uncodified local process innovations (Hurwicz's impossibility theorem).[14]

1.4 The Most Promising Architecture: Why We Reject a Global Blockchain

We explicitly reject building a per-transaction global blockchain (Architecture C), a centralized cloud database (Architecture A), or an unstructured gossip network without finality (Architecture B).

Participatory planning is not a stream of high-frequency financial trades; it is an epoch-synchronized, hierarchical bulk-synchronous coordination process followed by localized intra-year fulfillment. Furthermore, putting personal consumption requests on a public ledger violates personal privacy, and broadcasting 3.3 KB post-quantum signatures for every household action across all global nodes wastes bandwidth and energy.

Instead, our weighted evaluation (Chapter 6, scoring 4.56 / 5.00 vs. 2.47 / 5.00 for a blockchain) selects Architecture D — A Four-Layer Hybrid Local-First, Federated P2P, and Selective Epoch-BFT Architecture:

1. Layer 4 (Human Deliberation & Local Solver Edge): Accessible mobile/desktop Progressive Web App (PWA) and native clients running an embedded interior-point conic/quadratic solver (Clarabel.rs / HiGHS in WASM) for instant local proposal drafting and "what-if" collective-good exploration.
2. Layer 3 (Local-First Council & Household State): Encrypted local SQLite (SQLCipher) for structured vectors and accounting ledgers + Loro/Automerge CRDTs for offline-capable workplace and neighborhood council deliberation, authorized via revocable post-quantum capability tokens (PQ-UCAN).
3. Layer 2 (Federated Aggregation & Content-Addressed Data Plane): Peer-to-peer QUIC (quinn / iroh) transport organizing councils into nested Ward $\rightarrow$ City $\rightarrow$ Region and Industry Federation BLAKE3 Sparse Merkle-Sum Trees (MST), paired with content-addressed storage for qualitative council narratives (Lists).
4. Layer 1 (Selective Epoch-BFT Checkpoint & Deterministic IFB Engine): A lightweight rotating BFT validator committee (drawn from federated councils via Athenian-style cryptographic sortition) that commits only $\sim$10–20 global checkpoint blocks per annual planning cycle—anchoring each round's MST proposal root, executing the deterministic WASM price-update rule, and settling Sandström's macro Society Account.

1.5 Where AI Adds Value—and Where It Must Be Excluded

In strict adherence to Parecon's anti-coordinator principles,[15] AI must never become an unaccountable planning authority:

[15] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 15–17; Looking Forward, Ch. 9 (lf9_2.htm) and Ch. 10 (lf10_1.htm).

- 100% Deterministic Core: Indicative price adjustments, SB/SC ratios, and Society Account clearing use strictly deterministic fixed-point arithmetic and convex operations research (OR) solvers with zero neural-network weights.
- Accountable Edge Support: Classical statistical models (hierarchical ETS/ARIMA) forecast SKU size/color splits; local conic QP solvers (Clarabel.rs) automatically draft price-adjusted council revisions; and optional small local open-weight LLMs are restricted to multi-lingual translation, screen-reader accessibility, and summarizing qualitative Lists with mandatory links to raw council source text.

1.6 Estimated Effort, Budget Scenarios, and the Five Key Decisions

Table 1.2: Summary of Implementation Roadmap and Budget Scenarios (2026 Verified Rates)

Scenario	Staff & Time	Infra / Audit	Total Est. Budget & Deliverable
Scenario A Solo + AI Coding	1.0 FTE 4–6 mos	\$150–\$300 \$0 (self-fuzz)	\$50k–\$90k (or \$0 cash + 1 FTE): Open-source PoC, mobile PWA, 100k benchmark.
Scenario B (Rec.) Small Research Team	3.5–4.5 FTEs 6–9 mos	\$2.5k–\$5k \$25k–\$45k	\$380k–\$700k: Production PoC, 1M benchmarks, 50-person pilot.
Scenario C Funded Co-op Pilot	10–14 FTEs 24–36 mos	\$35k–\$75k \$120k–\$200k	\$2.4M–\$4.3M: Verified v1.0 protocol, 10–30 co-op field pilot.

The Five Most Important Decisions for Project Leadership

1. Commit to Architecture D (Local-First + Federated MST + Epoch Checkpoints) and Reject a Per-Transaction Blockchain.
2. Upgrade the Economic Solver from Unconstrained Cobb-Douglas to Constrained Conic/LP Optimization with Proximal Damping (Algorithm 3).
3. Enforce Hard Biophysical Caps ($r^*, E_z^{\max}$) Lexicographically Above PDRM Pollution Pricing.
4. Design for a "3-Milestone Hybrid Deliberation Schedule" separating automated intermediate input-clearing sub-iterations from human value-deliberation milestones.
5. Target Dual Utility: Support both full-scale Parecon simulation research and immediate deployment for federated worker cooperatives and participatory

budgeting communities ("Experiment 2: Developing a Parallel Economy").[16]

Final Recommendation: PROCEED immediately with the 16-week Proof-of-Concept and Non-Convex Convergence Benchmark Programme (Milestones M1–M5 in Chapter 9).

[16] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 129–130.

CHAPTER 2

Historical and Institutional Inspirations: From Council Practice to Protocol Design

> "Regular general assemblies of the syndicates or collectives were convened, in which all affairs were discussed and decided, and to which the administrative committees reported and remained at all times accountable and subject to recall. Self-management was not a slogan; it was the daily practice of millions of workers and peasants coordinating production in solidarity."
>
> — Gaston Leval, Collectives in the Spanish Revolution (1975) / Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (1974)[1]

2.1 Epistemological Premise: Institutions Before Cryptography

A recurring failure mode of contemporary "decentralized governance" (DAO) architectures is the conflation of cryptographic trustlessness with democratic self-management. Recording a tally on an immutable ledger does not render an economy democratic if:

1. Proposal drafting and agenda-setting are monopolized by an unaccountable technical or managerial elite—what Albert and Hahnel diagnose as the coordinator class.[2]
2. Decision-making influence is proportional to token wealth or meeting endurance rather than proportional to the degree participants are affected.
3. Wealthier production collectives hoard superior inherited productive endowments and refuse to equalize resources across regions.

In this chapter, we examine five historical and contemporary institutional traditions as empirical laboratories of democratic coordination—analyzing both their institutional breakthroughs and their structural failures—to derive binding architectural rules for DPEP.

[1] Gaston Leval, Collectives in the Spanish Revolution, trans. Vernon Richards (London: Freedom Press, 1975), Chs. 6–8; and Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), Chs. 2, 5, 9.

[2] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 14–17, 20–22; Looking Forward, Prologue ("Coordinatorism" and "The Origins of Coordinatorism").

2.2 The Spanish Revolution (1936–1939): Industrial and Agrarian Collectives

Following the defeat of the Nationalist military coup in July 1936 across Catalonia, Aragon, Valencia, and parts of Castile and Andalusia, workers and peasants affiliated with the anarchist CNT (Confederación Nacional del Trabajo) and socialist UGT (Unión General de Trabajadores) collectivized roughly 70–80% of industry in Catalonia and over 400 agrarian collectives in Aragon. Quantitative demographic studies by Frank Mintz place participation at a minimum of $1,838,000$ collectivists ($1,080,000$ industrial and $758,000$ agrarian, revised in 1977 from his earlier 1970 estimate of 2.44–3.2 million), while eyewitness researcher Gaston Leval estimated up to 3 million participants.[3]

2.2.1 Internal Self-Management, Delegation, and Recall

Across both Barcelona factories (tramways, textiles, metalworks, water/gas/electric utilities) and Aragonese villages (Alcañiz, Calanda, Binéfar, Graus, Mas de las Matas), supreme authority resided in the weekly or biweekly General Assembly (Asamblea General) of all workers or villagers.[4] Assemblies elected specialized administrative committees (Comités de Administración) to manage daily coordination, input procurement, and statistical reporting. As documented by Bookchin, Peirats, and Leval, these delegates operated under strict anti-bureaucratic constraints:

> "Where such popular assemblies were possible, their decisions were executed by a committee elected from the assembly. Apparently, the right to recall committee members was taken for granted and they certainly enjoyed no privileges, emoluments, or institutional power… It remained a cardinal principle of Spanish anarchists never to pay their delegates, even when the CNT numbered a million members. Normally, the responsibilities of elected delegates had to be discharged after working hours."[5]

2.2.2 Inter-Collective Coordination, Reorganization, and Accounting

- Regional Equalization in Aragon: Established in October 1936 (and formalized at the Congress of Caspe in February 1937 representing ∼450 collectives and $300,000$ collectivists), the Federation of Agrarian Collectives of Aragon pooled

[3] Frank Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, trans. Paul Sharkey (Oakland, CA: AK Press, 2013 [1970/1977]); Gaston Leval, Collectives in the Spanish Revolution (London: Freedom Press, 1975); Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), available in full at https://theanarchistlibrary.org/library/sam-dolgoff-editor-the-anarchist-collectives.

[4] Dolgoff, The Anarchist Collectives, Ch. 2 and Ch. 9; Murray Bookchin, "Introductory Essay," in Dolgoff, The Anarchist Collectives, pp. xxvii–xxx.

[5] Bookchin, "Introductory Essay," in Dolgoff, The Anarchist Collectives, p. xxviii.

surplus wheat, olive oil, and livestock at cantonal depots, transferred machinery and seed from fertile to arid villages, and conducted direct inter-regional trade with Catalonian textile syndicates without commercial middlemen.[6]

- Unified Export Coordination in Valencia (CLUEA): In the Levante, the CNT and UGT established the Consejo Levantino Unificado de Exportación Agrícola (CLUEA), federating ∼270 local orange and citrus committees to standardize grading, rail/maritime transport, and export clearing.[7]
- Industrial Recomposition in Catalonia: In Barcelona, the CNT's industrial unions (Sindicatos Únicos de Ramo) consolidated hundreds of small, unsanitary workshops in woodworking, baking, tanning, and optic/glass production into modern integrated plants, eliminating redundant middlemen while preserving full employment.[8]
- Accounting and the "Family Wage": Many Aragonese villages abolished official pesetas internally, distributing abundant staples free according to need (Maxim 4) and issuing a family consumption booklet (libreta de consumo) or local points calibrated to family size for scarce goods.[9] Inter-collective trade relied on standardized valuation tables (indexed in wheat equivalents or shadow pesetas) cleared through cantonal ledgers.

2.2.3 Historical Limitations and Structural Failures

Despite remarkable wartime productivity in munitions, transport, and agriculture, the Spanish collectives suffered three critical weaknesses that directly inform DPEP's design:

1. "Collective Egoism" (Neocapitalismo Colectivo) Between Rich and Poor Enterprises: In urban Catalonia, prior to the socialization decrees of late 1937, collectives remained financially autonomous inside a quasi-market environment. Wealthy collectives that inherited modern machinery or lucrative monopolies (such as the Barcelona tramways, power utilities, and luxury hotels) enjoyed large surpluses, while raw-material-starved textile and construction collectives struggled to pay base wages.[10] Protocol Lesson: This historical failure proves why Albert and Hahnel (1991a, Ch. 5, pp. 84–87) and Sandström (2020) insist that productive assets belong to the Productive Commons, and that every Worker Council must be charged an opportunity-cost rental fee ($P_k \cdot k_j$) on its inherited capital stock!

[6] Leval, Collectives in the Spanish Revolution, Ch. 6; Dolgoff, The Anarchist Collectives, Ch. 10.
[7] Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, Ch. 4.
[8] Augustin Souchy, "Collectivizations in Catalonia," in Dolgoff, The Anarchist Collectives, Ch. 7.
[9] Leval, Collectives in the Spanish Revolution, pp. 165–210.
[10] José Peirats, "Pay Scales and Collectible Egoism," in Dolgoff, The Anarchist Collectives, Ch. 6; Mintz, Anarchism and Workers' Self-Management.

2. Burnout from Unpaid Evening Governance: Requiring delegates to perform all federation administration after a full day of physical factory labor caused severe exhaustion and favored childless militants over workers with caregiving duties. Protocol Lesson: Governance and facilitation work must be counted inside paid work hours as part of every citizen's Balanced Job Complex.
3. Lack of Unified Shadow Pricing for Complex Intermediate Chains: While cantonal barter cleared simple agricultural staples, urban industries lacked an iterative mechanism to compute social opportunity costs across hundreds of interdependent intermediate inputs once pre-war market prices grew stale.

2.3 The Paris Commune (March 18 – May 28, 1871)

During its 72 days of existence before its bloody suppression during La Semaine Sanglante, the Paris Commune established foundational principles of council democracy celebrated in Karl Marx's The Civil War in France (1871).[11] A rigorous institutional analysis must distinguish what the Commune actually implemented from unimplemented proposals and later theoretical extrapolations.

2.4 Classical Athenian Democracy (∼508/7 – 322 BCE)

Over nearly two centuries from the reforms of Kleisthenes (508/7 BCE) to the Macedonian conquest (322 BCE), classical Athens developed the most sophisticated institutional machinery for direct citizen self-governance in antiquity, analyzed definitively by Mogens Herman Hansen.[14]

2.4.1 Core Institutional Mechanisms Adaptable to DPEP

1. Direct Assembly (Ekklesia) vs. Norm-Setting (Nomothetai): Open to all adult male citizens (∼30,000–40,000 eligible in the 4th century BCE, with ∼6,000 attending the 40 annual meetings on the Pnyx), the Ekklesia voted on decrees (psephismata), while permanent constitutional laws (nomoi) were ratified by a separate panel of 1,000 nomothetai selected by lot.
2. Sortition (Kleroterion) and Strict Rotation: Roughly 600 of the ∼700 domestic magistrates (archai) and the 6,000 annual jurors (dikastai) were selected by lot using bronze allotment machines (kleroteria) rather than election—following Aristotle's principle (Politics 1294b) that elections are intrinsically oligarchic (favoring the wealthy, famous, and rhetorically polished) whereas

[11] Karl Marx, The Civil War in France (London: General Council of the International Workingmen's Association, 1871); John Merriman, Massacre: The Life and Death of the Paris Commune of 1871 (New York: Basic Books, 2014).
[12] Marx, The Civil War in France, Section III.
[13] Merriman, Massacre: The Life and Death of the Paris Commune of 1871, Ch. 5.
[14] Mogens Herman Hansen, The Athenian Democracy in the Age of Demosthenes: Structure, Principles, and Ideology (Oxford: Blackwell / Norman: University of Oklahoma Press, 1991).

Table 2.1: Paris Commune (1871): Actually Implemented Institutions vs. Proposals & Extrapolations

Dimension	Actually Implemented (March–May 1871)	Proposed on Paper or Later Extrapolation
Representation & Recall	Municipal elections across 20 arrondissements (March 26, 1871) by universal male suffrage; councilors declared revocable at any time; merged legislative and executive work across 10 Commissions.[12]	Formal recall was rarely exercised during the 72-day military siege; women were excluded from formal voting despite leading grass-roots defense and cooperative organizing via the Union des Femmes (Élisabeth Dmitrieff, Nathalie Lemel).
Remuneration Cap	Decree of April 2, 1871: Capped maximum salaries of all Commune officials and municipal administrators at 6,000 francs/year (equivalent to a skilled Parisian workman's wage) and banned multiple stipends.	Did not abolish wage hierarchies inside private Parisian enterprises or establish balanced job complexes inside workshops.
Workplace Self-Management	Decree of April 16, 1871 (drafted by Léo Frankel's Commission of Labor and Exchange): Ordered a census of workshops abandoned by fleeing owners and their transfer to worker cooperative associations; abolished night baking (April 20) and employer wage fines (April 27).[13]	Only a few dozen cooperative workshops (e.g., the Louvre armaments workshop electing its own council and foremen; women's uniform cooperatives) actually began operating before Versailles troops entered Paris on May 21.
National Federation & Credit	Operated strictly within besieged Paris; left the Banque de France intact (negotiating advances via Charles Beslay rather than socializing credit).	Declaration to the French People (April 19, 1871) (drafted by Proudhonian Pierre Denis) and Marx envisioned a national federation of autonomous communes coordinating via delegated national assemblies—never realized due to the crushing of provincial communes in Lyon and Marseille.
Crisis Centralization	On May 1, 1871, under bombardment, the Jacobin/Blanquist majority voted (45 to 23) to create a 5-member Committee of Public Safety with sweeping powers over the objection of the Federalist minority (Frankel, Varlin, Longuet, Courbet).	Demonstrates the acute vulnerability of council democracies during crisis to revert to emergency command authority unless crisis-mode bounds (TTLs and post-incident audits) are constitutionally hard-coded.

sortition is democratic.[15] Almost all offices were restricted to a non-repeatable one-year term (or repeatable only once in a lifetime for the Boule), preventing a permanent coordinator class.

3. Probouleutic Agenda Facilitation (Boule of 500): Five hundred citizens (50 from each of the 10 Kleisthenic tribes, chosen by lot from local demes) served for one year on the Council (Boule), with each tribe serving as the standing executive committee (prytaneis) for 1/10th of the year and a new chair (epistates) drawn by lot every 24 hours. Crucially, the Boule's role was probouleutic: it prepared structured proposals (probouleumata) for the Ekklesia to debate, amend, or reject, but could not impose major policies on its own.[16] This is the direct institutional ancestor of Parecon's Facilitation Boards.

4. Public Scrutiny and Mandatory Audit (Dokimasia, Euthynai, Graphe Paranomon): Every magistrate underwent pre-office eligibility scrutiny (dokimasia) and a mandatory end-of-term financial and conduct audit (euthynai) before 10 logistai (accountants) and 10 euthynoi chosen by lot.[17]

5. Civic Remuneration (Misthophoria): Introduced by Pericles for jurors and councilors and expanded in the 4th century BCE to assembly attendance (misthos ekklesiastikos, 3 to 6 obols per day), ensuring working citizens (thetes) could afford to participate.

2.4.2 Explicit Examination of Structural Exclusions

Athenian democracy excluded women (who lacked political citizenship and independent property control), metics (resident non-citizens, ∼20,000–30,000 artisans and merchants), and enslaved people (∼80,000–150,000 forced laborers in the Laurion silver mines, agriculture, crafts, and households), leaving only ∼10–15% of the total population with full political rights.[18] In adapting Athenian mechanisms to a universal participatory economy, DPEP replaces citizenship exclusion and coerced labor with: (a) universal adult council membership across all residents and workers, (b) Balanced Job Complexes that distribute rote and empowering work equitably across everyone, (c) accounting for governance/audit participation inside paid BJC work hours (universalizing misthophoria), and (d) cryptographic Verifiable Random Function (VRF) sortition to staff IFB oversight boards, industry innovation juries, and audit panels.

[15] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 9, pp. 230–245.
[16] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 10, pp. 246–265.
[17] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 9, pp. 218–224.
[18] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 4, pp. 86–124.

2.5 Occupy Wall Street (2011) and Additional Precedents

- Occupy Wall Street and Horizontal Movements (2011): General Assemblies in Zuccotti Park and Oakland used hand signals, "progressive stack," working groups, and spokescouncils under modified consensus (90% supermajority).[19] While effective at building initial solidarity, plenary assemblies suffered severe $\mathcal{O}(N^2)$ synchronous speaking bottlenecks, block paralysis, attrition bias, and informal cliques. In Parecon: Life After Capitalism (2003, Ch. 5, "Consensus?"), Michael Albert anticipates this exact dilemma by distinguishing participatory preparation (open deliberation and information sharing) from unanimity veto rules: unanimity works well when hiring a teammate into a small work group (where one member's severe discomfort outweighs others' mild preference), whereas plant-wide investment choices are better served by majority/supermajority voting paired with a two-step minority delay option.[20]
- Mondragón Cooperative Federation (1956–Present): Demonstrates across $> 80,000$ worker-owners in $\sim$90 cooperatives that one-worker-one-vote assemblies, Watchdog Audit Councils (Consejo Social), and inter-cooperative solidarity funds can operate complex industrial manufacturing at scale,[21] while confirming Albert and Hahnel's thesis (1991a, Ch. 1, p. 14) that external market competition pressures internal wage ratios unless replaced by participatory allocation.
- Porto Alegre Participatory Budgeting (1989–Present): Combined neighborhood assemblies and recallable delegates with an objective Quality of Life Need Matrix (weighting lack of water, sewer, pavement, and population), successfully redirecting municipal capital investment to poorer districts.
- Rojava / Democratic Confederalism (2012–Present): Implemented nested communes with a strict co-chair gender parity rule and autonomous women's councils (Kongra Star) holding veto power over gender-affecting decisions—paralleling Hahnel, Bohmer, and Chowdhury's proposal for Women's Caucuses in Parecon.[22]
- Ostrom's Common-Pool Resource (CPR) Institutions (1990): Elinor Ostrom's empirical study of enduring self-governed commons (such as Valencia's Tribunal de las Aguas) established eight design principles: clearly defined boundaries, proportional equivalence of benefits and costs, collective-choice arrangements, accountable monitoring, graduated sanctions, rapid local conflict reso-

[19] Graeber, The Democracy Project, Chs. 1–3; Freeman, "The Tyranny of Structurelessness," pp. 151–164.

[20] Michael Albert, Parecon: Life After Capitalism (Verso, 2003), Ch. 5 ("Councils: Consensus?"), pp. 93–100.

[21] William Foote Whyte and Kathleen King Whyte, Making Mondragón: The Growth and Dynamics of the Worker Cooperative Complex, 2nd ed. (Ithaca, NY: ILR Press, 1991).

[22] Hahnel, Bohmer, and Chowdhury, "Reproductive Labor," Ch. 10 in Hahnel, Democratic Economic Planning, p. 200.

lution, right to organize, and nested enterprises.[23]

[23] Elinor Ostrom, Governing the Commons: The Evolution of Institutions for Collective Action (Cambridge: Cambridge University Press, 1990), Ch. 3, p. 90.

Synthesis: Comparative Institutional-to-Protocol Matrix

Table 2.2 synthesizes the eight core institutional mechanisms analyzed across Chapters 2.2–2.5, mapping their historical strengths and limitations directly to concrete capabilities in the Distributed Participatory Economics Protocol (DPEP).

Table 2.2: Comparative Matrix: Institutional Mechanisms Mapped to DPEP Protocol Capabilities

Mechanism & Precedents	Strengths & Advantages	Historical Limitations	Concrete DPEP Protocol Capability
1. General Assembly Spain 1936; Athens Ekklesia; Occupy; Mondragón	Builds face-to-face solidarity; surfaces qualitative context.	Fails above ∼50–150 speakers ($\mathcal{O}(N^2)$ friction); meeting fatigue.	Scoped strictly to work teams (10–30) and neighborhood councils (50–200) with async pre-reading queues.
2. Revocable Delegation Paris Commune 1871; Spain CNT; Porto Alegre PB	Enables multi-tier federation scaling while tethering delegates to base.	Delegates ossify if recall is procedurally slow or difficult.	Scoped PQ-UCAN Delegate Certificates: Issued by council threshold sig; revocable in a single council vote.
3. Sortition + Rotation Classical Athens (Boule, Logistai, Dikasteria)	Prevents factional elections, charismatic capture, and coordinator elites.	Unsuitable for rare specialist tasks unless drawn from qualified pool.	VRF Cryptographic Sortition: Selects rotating IFB oversight, audit panels, and dispute juries.
4. Mandatory Audit Athens Euthynai (Logistai); Mondragón Consejo Social	Systematically detects bias or unauthorized overrides.	Requires tamper-evident records and independent auditors.	Deterministic WASM + Merkle-Sum Audit Logs: Any node can verify IFB price updates in < 10 ms.
5. Endowment Equalization Aragon Depots (1936); Mondragón Funds	Prevents collectives with superior machinery from extracting rents.	Failed in 1936 Barcelona where rich collectives hoarded revenues.	Stand-Alone Capital Asset Ledger: Automatically debits each WC_j's SC_j for capital/resource user rights (P_k, P_r).
6. Proportional & Caucus Rojava Co-Chairs & Women's Councils	Protects disproportionately affected groups from majority tyranny.	Can deadlock if scope of veto is unbounded.	Impact-Weighted & Dual-Key Governance Rules (PEPE Ch. 2 p. 24; DEP Ch. 10 p. 200).
7. Iterative Tâtonnement IETF/Git; Lange-Arrow-Hurwicz; Parecon IFB	Scales quantitative clearing to millions asynchronously.	Pure prices strip out human context unless paired with qualitative text.	Federated Sparse Merkle-Sum Trees + Content-Addressed Qualitative Lists.
8. Graduated Sanctions Ostrom CPRs (Valencia Tribunal de las Aguas)	Resolves 95% of operational friction locally with minimal overhead.	Requires clear invariants and fast local mediation.	Tiered Variance Alerts → UFB Mediation → Industry/Environmental Court.

CHAPTER 3

Source-Grounded Requirements of Participatory Economics

> "Until now, however, descriptions of alternatives have been unconvincing. Here Michael Albert and Robin Hahnel support the libertarian socialist tradition by presenting a rigorous, well-defined model of how producers and consumers could democratically plan their interconnected activities."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Front Flap / JSTOR Abstract[1]

3.1 Source Provenance and Classification Methodology

Before formulating mathematical equations or distributed protocols, we establish strict traceability to the foundational texts. Every requirement in Table 3.1 is classified into one of four provenance tiers:

- [ORIGINAL-1991]: Explicitly specified in Michael Albert and Robin Hahnel's two 1991 volumes, The Political Economy of Participatory Economics (PEPE) and Looking Forward (LF).[2]
- [INTERPRETATION-LATER]: Institutional elaborations, formalizations, and accounting procedures introduced in Michael Albert's Parecon: Life After Capitalism (PLAC, 2003), Robin Hahnel's Democratic Economic Planning (DEP, 2021) and A Participatory Economy (2022), Anders Sandström's Anarchist Accounting (2020/2023), Mitchell Szczepanczyk's simulation work (2023), and the Participatory Economy Project's User Story Mapping (2023).[3]
- [TECH-EXTENSION]: Technological extensions required to execute the model over a modern, decentralized, post-quantum, privacy-preserving computational network.
- [UNRESOLVED]: Open theoretical or institutional questions requiring simulation or empirical human research (examined in full in Chapter 10).

[1] Albert and Hahnel, Political Economy of Participatory Economics, book overview (https://www.jstor.org/stable/j.ctv131bvr2).

[2] Albert and Hahnel, Political Economy of Participatory Economics (1991), print pagination verified via library TOC scan (research/sources/pepe_dandelon.pdf, pp. 1–132) and full text analyzed via ZNetwork (intro.htm, 1.htm–6.htm); Albert and Hahnel, Looking Forward (1991), full text analyzed via ZNetwork (lfIntroduction.htm, lf1_1.htm–lf10_1.htm).

[3] Hahnel, Democratic Economic Planning (2021); Robin Hahnel, A Participatory Economy (Chico, CA: AK Press, 2022); Sandström, Anarchist Accounting (2020); Szczepanczyk, "Pseudocode and Algorithms" (2023); Participatory Economy Project, "Participatory Planning: User Story Mapping" (2023), https://participatoryeconomy.org/project/participatory-planning-user-story-mapping/.

3.2 Normative Foundations and Endogenous Preference Theorems

In Chapter 1 of The Political Economy of Participatory Economics (pp. 8–11), Albert and Hahnel evaluate economic institutions against five valuative criteria: Equity, Self-Management, Solidarity, Variety, and Efficiency.[4] For distributive equity, they distinguish four mutually exclusive maxims (PEPE Ch. 1, p. 8; Ch. 3, p. 44; DEP Ch. 1, pp. 16–22):

1. Maxim 1 (Capitalism): Payment according to the value of the contribution of one's personal labor and productive property owned (rejected due to inheritance—the "Rockefeller grandson problem"—luck, and compounding wealth accumulation).[5]
2. Maxim 2 (Market Socialism / Coordinatorism): Payment according to the value of one's personal labor contribution alone (rejected because differences in marginal productivity stem overwhelmingly from the "genetic lottery," luck, job assignment, and publicly funded training rather than personal sacrifice—illustrated by their "Mozart vs. Salieri" and "Dave Butz" examples).[6]
3. Maxim 3 (Participatory Economics): Payment according to effort and personal sacrifice expended in socially valued labor (longer hours, higher intensity, or more onerous/dangerous conditions prior to full BJC equalization).
4. Maxim 4 (Humane Solidarity): Payment according to need—incorporated into Parecon via universal free public goods (healthcare, education) and council-approved special-need allowances (PEPE Ch. 3, p. 47).

Crucially, building on their mathematical proofs in Quiet Revolution in Welfare Economics (1990),[7] Albert and Hahnel prove that because human preferences are endogenous (people rationally mold their future characteristics ΔC in anticipation of future terms of supply), any institution that systematically mischarges for activities relative to true social opportunity cost causes: (1) a snowballing divergence from Pareto optimality over time, (2) systematic warping of human development trajectories, and (3) ex post invisibility of the bias to participants. Markets systematically underprice negative externalities and overprice public goods (warping preferences toward anti-social individualism), while central planning systematically under-supplies self-managed work roles (warping workers toward apathy and coordinator submission).[8]

[4] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 8–9, 11.
[5] Hahnel, Democratic Economic Planning, Ch. 1, pp. 16–18; Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 45–46.
[6] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 46–47.
[7] Robin Hahnel and Michael Albert, Quiet Revolution in Welfare Economics (Princeton, NJ: Princeton University Press, 1990), Chs. 4, 6, 7; summarized in Political Economy of Participatory Economics, Ch. 1, pp. 9–11.
[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 12–17; Ch. 5, pp. 98–105.

3.3 Formal Traceability Matrix of Requirements

Table 3.1: Master Requirements Matrix Derived from Primary and Secondary Literature

Req ID	Requirement Specification	Classification	Primary Citations
REQ-NORM-01	Productive Commons & User-Right Fees: Abolish private ownership of productive property; all natural resources and capital stocks belong to society. Worker councils acquire temporary user rights during annual planning and pay opportunity-cost charges (P_r, P_k, P_l).	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 1 (pp. 17–20), Ch. 5 (p. 85); DEP Ch. 6 (pp. 98–102), Ch. 12 (p. 244); Sandström (2020).
REQ-NORM-02	Proportional Self-Management: Decision-making input proportional to the degree affected by an economic choice.	[ORIGINAL-1991]	PEPE Ch. 1 (p. 8), Ch. 2 (p. 24); LF Ch. 1; PLAC Ch. 5; DEP Ch. 1 (p. 14).
REQ-NORM-03	Effort & Sacrifice Remuneration: Consumption budgets proportional to effort/sacrifice (Maxim 3) tempered by universal free goods and need exceptions (Maxim 4), with council average effort ratings capped by SB_j/SC_j.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 3 (pp. 44–55); DEP Ch. 6 (pp. 108–117); Sandström (2023).
REQ-NORM-04	Balanced Job Complexes (BJCs): Tasks partitioned into subsets $\{PIPA\}_i$ ordered by empowerment/desirability and combined into balanced complexes within and across workplaces via Job Complex Committees.	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 25–30), Ch. 5 (pp. 82–83); LF Chs. 1–2.
REQ-ACT-01	Nested Consumer Councils & Federations: Individuals/households → Neighborhood CC_h → Ward → City/County → State/Region → National Federation, requesting public goods matched to spatial scale.	[ORIGINAL-1991]	PEPE Ch. 3 (pp. 40–42), Ch. 5 (pp. 81–82); LF Chs. 3, 4, 8.
REQ-ACT-02	Worker Councils & Dual Federations: Democratic workplace councils federated both by industry (scale economies, R&D, SB/SC benchmarking) and by region (local production externalities).	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 23–25), Ch. 4 (p. 62), Ch. 5 (p. 88); LF Chs. 1, 2, 7.
REQ-ACT-03	Six Functional Facilitation Boards: Iteration (IFB), Consumption (CFB), Production (PFB), Employment (EFB), Household (HFB), and Updating (UFB) facilitation boards performing purely informational/mediating tasks.	[ORIGINAL-1991]	LF Ch. 6 (lf6_1.htm), Ch. 9 (lf9_2.htm); PEPE Ch. 4 (p. 62).
REQ-ACT-04	Communities of Affected Parties (CAPs) & Environmental Courts: Airshed/watershed collectives participating in the Pollution Demand-Revealing Mechanism (PDRM) below hard biophysical caps.	[INTERP-LATER]	DEP Ch. 7 (pp. 138–149), Ch. 14 (pp. 261–267); PEP (2023).

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Req ID	Requirement Specification	Classification	Primary Citations
REQ-ACT-05	Stand-Alone Capital Asset Entities & Society Account: Double-entry social accounting entities tracking capital production costs, user-right fees, CAP flows, and GDP division between consumption and investment.	[INTERP-LATER]	Sandström (2020, Chs. 3–5; 2023).
REQ-DATA-01	Seven Priced Economic Categories: Private goods, public goods, intermediate inputs, natural resources, labor categories, capital stocks, and regulated pollutants.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 5 (pp. 78–88); DEP Ch. 7, Ch. 9; Szczepanczyk (2023).
REQ-DATA-02	Coarse Planning Taxonomy vs. Granular SKUs: Plan over coarse resource-equivalent product classes; extrapolate specific sizes/colors statistically and resolve at intra-year fulfillment.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 4 (p. 64); LF Ch. 6 (lf6_2.htm); DEP Ch. 8 (pp. 163–167).
REQ-DATA-03	Public Books of Blueprints & Qualitative Lists: Publish all technical input-output recipes $[A, K, R, L]$ and attach qualitative descriptive Lists to quantitative council vectors.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 61–62), Ch. 5 (p. 84), Ch. 6; PLAC Chs. 11, 17.
REQ-DATA-04	Protected Private Consumption Sphere: Personal consumption requests within warranted budget B_u cannot be vetoed by neighbors and must be cryptographically shielded from neighbor snooping.	[ORIGINAL-1991] + [TECH-EXT]	PEPE Ch. 3 (p. 41), Ch. 5 (p. 80); LF Ch. 3; PLAC Chs. 12, 18; Dapprich (2024).
REQ-PLAN-01	Multi-Horizon & Collective-First Sequencing: Long-run (education, environment) and aggregate investment plans precede annual planning; federation collective goods are settled before final private consumption bundles.	[ORIGINAL-1991] + [INTERP-LATER]	LF Ch. 3 (lf3_2.htm), Ch. 8 (lf8_1.htm), Ch. 9 (lf9_1.htm); DEP Chs. 11–15.
REQ-PLAN-02	Slack Planning & Hierarchical Intra-Year Clearing: Include 3–5% planned excess supply buffer (SL^E); cancel routine consumer swaps inside neighborhood/ward/city federations; trigger UFB replanning only when net shocks exceed slack.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 66–67), Ch. 6 (p. 113, 122); DEP Ch. 8 (pp. 167–170).

CHAPTER 4

Formal Economic Model, Accounting Identities, and Algorithms

> "It is impossible to show that a feasible nonmarket system at least approaches the productivity of the market unless (1) a rather well-developed theoretical model of the nonmarket system is available, and (2) it is demonstrated that a sufficiently productive approximation of the ideal… system described in the theoretical model is practically possible."
>
> — Allen Buchanan, Ethics, Efficiency, and the Market (1985), p. 29, quoted in Albert and Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 58, and Ch. 5, p. 73[1]

4.1 Ontology of Economic Actors and Priced Objects

Let the participatory economy $\mathcal{E}$ over annual planning period T consist of the following finite sets of actors and institutional entities:[2]

- Individuals ($\mathcal{I} = \{1, \ldots, N_{\text{ind}}\}$): Natural persons participating as consumers, workers, and citizens. Each individual $u \in \mathcal{I}$ belongs to a primary neighborhood consumer council $h(u) \in \mathcal{H}$ and, if working, a primary worker council $j(u) \in \mathcal{J}$ (plus optional secondary council $j'(u)$ for cross-workplace Balanced Job Complex rotation).[3]
- Neighborhood Consumer Councils ($\mathcal{H} = \{CC_1, \ldots, CC_H\}$) & Nested Consumer Federations ($\mathcal{F}^C = \bigcup_{\ell=1}^{L} \mathcal{F}_\ell^C$): A rooted geographic hierarchy of depth $L = 5$ (Level 1 = Ward, Level 2 = City/County, Level 3 = State, Level 4 = Region, Level 5 = National Federation).[4]
- Worker Councils ($\mathcal{J} = \{WC_1, \ldots, WC_J\}$) & Dual Worker Federations ($\mathcal{F}^W = \mathcal{F}^{W,\text{ind}} \cup \mathcal{F}^{W,\text{reg}}$): Democratic workplace collectives federated both by industry ($IF_k \in \mathcal{F}^{W,\text{ind}}$) and by geographic region ($\mathcal{F}^{W,\text{reg}}$).[5]

[1] Allen Buchanan, Ethics, Efficiency, and the Market (Totowa, NJ: Rowman and Littlefield, 1985), p. 29; cited in Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, p. 58 n. 3, and Ch. 5, p. 73.

[2] Formalized from Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4 (pp. 62–68) and Ch. 5 (pp. 77–95); Hahnel, Democratic Economic Planning, Ch. 7 (pp. 130–156); Sandström, Anarchist Accounting (2020), Ch. 3; and Participatory Economy Project, "User Story Mapping" (2023).

[3] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 27–30; Ch. 5, p. 83.

[4] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 40–42; Ch. 5, pp. 81–82.

[5] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 23–25; Ch. 5, pp. 88, 96–97.

- Communities of Affected Parties ($\mathcal{C}^{\mathrm{CAP}} = \{CAP_1, \ldots, CAP_Z\}$): Ecological collectives of residents $\mathrm{pop}(CAP_z) \subseteq \mathcal{I}$ impacted by pollutant $z \in \mathcal{Z}$ in a specific airshed or watershed.[6]
- Six Facilitation Boards ($\mathcal{B} = \{\mathrm{IFB}, \mathrm{CFB}, \mathrm{PFB}, \mathrm{EFB}, \mathrm{HFB}, \mathrm{UFB}\}$), Stand-Alone Capital Asset Entities ($\mathcal{A}^{\mathrm{cap}}$), and the Society Account ($\mathcal{A}^{\mathrm{soc}}$).[7]

Definition 4.1 (Master Catalog of Seven Priced Economic Categories). Following Albert and Hahnel (1991a, Chs. 5–6), Hahnel (2021, Chs. 7–9), and Szczepanczyk (2023, Sec. 4), the economy tracks supply, demand, and indicative prices across seven disjoint categories indexed by the master catalog Ω:

$$\Omega = \mathcal{G}^{\mathrm{priv}} \sqcup \mathcal{G}^{\mathrm{pub}} \sqcup \mathcal{G}^{\mathrm{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z} \tag{4.1}$$

with complete indicative price vector at iteration t given by:

$$P^{(t)} = \left(p^{\mathrm{priv}},\, p^{\mathrm{pub}},\, p^{\mathrm{int}},\, p^{\mathrm{res}},\, p^{\mathrm{lab}},\, p^{\mathrm{cap}},\, p^{\mathrm{pol}}\right) \in \mathbb{R}_{+}^{|\Omega|} \tag{4.2}$$

4.2 Primal-Dual Reference Model and Sandström's Society Account

In Chapter 5 of The Political Economy of Participatory Economics (pp. 78–79), Albert and Hahnel introduce a primal-dual linear programming reference model to clarify the meaning of shadow prices before presenting their decentralized iterative procedure.[8] Let x be the vector of production activity levels, A be the produced intermediate input matrix, K be the capital stock coefficient matrix (with available stock k^*), R be the non-producible natural resource input matrix (with available ecological supply r^*), L be the direct labor input matrix (with available labor l^*), and v be the row vector of relative social values of net outputs. The economy-wide primal and dual reference problems are:

$$\text{Primal:} \quad \max_{x \geq 0} \; v(I - A)x \quad \text{subject to} \quad Kx \leq k^*,\; Rx \leq r^*,\; Lx \leq l^* \tag{4.3}$$

$$\text{Dual:} \quad \min_{P_k, P_r, P_l \geq 0} \; P_k k^* + P_r r^* + P_l l^* \quad \text{subject to} \quad P_k K + P_r R + P_l L \geq v(I - A) \tag{4.4}$$

As Albert and Hahnel explicitly note (PEPE Ch. 5, p. 79), the shadow prices P_l solved from the static dual (4.4) reflect only the marginal physical productivities of different labor categories, not the differential disutilities or desirabilities of work from the workers' point of view; whereas the indicative prices generated by Parecon's iterative council procedure incorporate worker well-being and effort disutility directly.[9]

[6] Hahnel, Democratic Economic Planning, Ch. 7 ("The Pollution Demand Revealing Mechanism"), pp. 139–148.

[7] Albert and Hahnel, Looking Forward, Ch. 6 (lf6_1.htm); Sandström, "An Accounting System for a Libertarian Socialist Economy" (2023).

[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 78–79.

[9] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, p. 79.

4.2.1 Closing the Macro-Accounting Loop (Sandström 2020/2023)

Anders Sandström's Anarchist Accounting (2020) and 2023 technical specification formalize how opportunity-cost user fees paid by Worker Councils reconcile with effort-based consumer incomes inside the Society Account ($\mathcal{A}^{\text{soc}}$):[10]

1. Social Value of Net Final Output (GDP) at Indicative Prices P:

$$Y^{\text{GDP}}(P) = p^{\text{priv}} \cdot D^{\text{priv}} + p^{\text{pub}} \cdot D^{\text{pub}} + p^{\text{cap,new}} \cdot D^{\text{inv}} \tag{4.5}$$

   In the Society Account, credits consist of all user-right fees charged to Worker Councils for access to society's labor categories ($p^{\text{lab}} \cdot L$), natural resources ($p^{\text{res}} \cdot R$), and productive capital assets held in Stand-Alone Capital Asset Entities ($p^{\text{cap}} \cdot K$), plus Consumer Council depreciation/discount charges on public consumer assets and net WC operating surpluses.

2. Aggregate Consumption Pool and Base Hourly Remuneration (w_{base}): Let $\sigma_{\text{inv}} \in (0,1)$ be the share of GDP allocated to aggregate investment (decided in Aggregate Investment Planning; DEP Ch. 11),[11] and let Y^{allow} be the politically ratified budget for non-working allowances (children, retirees, disability) and universal free public services (healthcare, education; Maxim 4). For total planned socially valued work hours $H^{\text{tot}} = \sum_{u \in \mathcal{I}^{\text{work}}} h_u$, the base remuneration per standard effort-adjusted hour is:

$$Y^{\text{work}}(P) = (1 - \sigma_{\text{inv}}) Y^{\text{GDP}}(P) - Y^{\text{allow}}, \qquad w_{\text{base}}(P) = \frac{Y^{\text{work}}(P)}{H^{\text{tot}}} \tag{4.6}$$

3. Capping Workplace Effort Ratings by SB_j/SC_j (DEP Ch. 6, p. 117): To prevent collective "grade inflation" where a worker council awards all its members inflated effort multipliers ϵ_u, Hahnel (2021, Ch. 6, p. 117) caps each worker council's hours-weighted average effort rating by its social benefit-to-cost ratio:[12]

$$\frac{\sum_{u \in WC_j} h_u \epsilon_u}{\sum_{u \in WC_j} h_u} \leq \min\left(\bar{\epsilon}_{\max}, \frac{SB_j(P)}{SC_j(P)}\right) \tag{4.7}$$

4. Individual Warranted Annual Consumption Budget (B_u): Combining Sandström's four income flows (Sandström 2023, Arrows 1–4), citizen u's warranted budget for private and collective consumption is:

$$B_u(P) = \underbrace{w_{\text{base}}(P) h_u \epsilon_u}_{\text{1: Work Effort}} + \underbrace{A_u^{\text{social}}}_{\text{2: Benefits}} + \underbrace{\sum_{z: u \in CAP_z} \frac{p_z^{\text{pol}} E_z}{|\text{pop}(CAP_z)|}}_{\text{3: CAP Compensation}} + \underbrace{\Delta S_u}_{\text{4: Loan/Need}} \tag{4.8}$$

[10] Sandström, Anarchist Accounting (2020), Ch. 3; Sandström, "An Accounting System for a Libertarian Socialist Economy" (2023), Diagram 1 (Arrows 1–14).

[11] Hahnel and Kerkhoff, "Aggregate Investment Planning," Ch. 11 in Hahnel, Democratic Economic Planning, pp. 219–237.

[12] Hahnel, Democratic Economic Planning, Ch. 6 ("Capping average effort ratings"), p. 117.

4.3 Formal Optimization Problems of Councils and CAPs

4.3.1 Consumer Councils and Nested Federations (PEPE Ch. 5, pp. 79–82, 88)

Each consumer council $CC_h \in \mathcal{H}$ seeks to maximize its members' total economic well-being $W_h(U_h, \Delta C_h)$—incorporating both immediate preference fulfillment U_h and endogenous human characteristic development ΔC_h—subject to the equitable budget constraint that its total social cost $SC(CC_h, P)$ does not exceed its members' warranted budget $B(CC_h, P) = \sum_{u \in \mathrm{pop}(CC_h)} B_u(P)$:[13]

$$\max_{y_h^{\mathrm{priv}},\, y_h^{\mathrm{pub,loc}} \geq 0} \quad W_h\Big(U_h(y_h^{\mathrm{priv}}, y^{\mathrm{pub}}),\ \Delta C_h(y_h^{\mathrm{priv}}, y^{\mathrm{pub}})\Big) \tag{4.9}$$

$$\text{subject to} \quad SC(CC_h, P) \triangleq p^{\mathrm{priv}} \cdot y_h^{\mathrm{priv}} + p^{\mathrm{pub}} \cdot y_h^{\mathrm{pub,loc}} + |\mathrm{pop}(CC_h)| \sum_{f \succ CC_h} \frac{p^{\mathrm{pub}} \cdot y_f^{\mathrm{pub}}}{|\mathrm{pop}(f)|} \leq B(C$$

where $f \succ CC_h$ denotes every higher-level ward, city, state, or national consumer federation containing CC_h. As proven in PEPE Ch. 3 (p. 41) and Ch. 5 (p. 80), for any individual or council whose request satisfies $SC \leq B$, other councils may advise but cannot veto the request; veto power applies only to exception requests exceeding warranted allowances ($SC > B$).

4.3.2 Communities of Affected Parties (CAPs) (DEP Ch. 7, pp. 139–148)

For each regulated pollutant $z \in \mathcal{Z}$ with hard biophysical sustainability ceiling $E_z^{\max}$ (established by Long-Run Environmental Planning; DEP Ch. 14),[14] given indicative damage price p_z^{pol}, CAP_z chooses the total permissive emission supply $E_z \in [0, E_z^{\max}]$ to maximize member well-being:

$$\max_{0 \leq E_z \leq E_z^{\max}} \quad W_{CAP_z}\Big(p_z^{\mathrm{pol}} E_z,\ -D_{CAP_z}(E_z)\Big) \tag{4.10}$$

At an interior optimum below $E_z^{\max}$, CAP_z sets E_z such that the sum of marginal victim damages across $u \in \mathrm{pop}(CAP_z)$ equals p_z^{pol} (the Lindahl-Pigou efficiency condition).

4.3.3 Worker Councils (PEPE Ch. 5, pp. 85–88; DEP Ch. 7, pp. 152–156)

Each worker council $WC_j \in \mathcal{J}$ begins the planning cycle with initial labor endowment l_j^0 ($\sum_{j=1}^{J} l_j^0 = l^*$) and initial capital stock endowment k_j^0 ($\sum_{j=1}^{J} k_j^0 = k^*$), for which it is automatically charged $p^{\mathrm{lab}} \cdot l_j^0 + p^{\mathrm{cap}} \cdot k_j^0$.[15] In iteration t, WC_j proposes output vector $\bar{a}_j = (\bar{a}_j^{\mathrm{priv}}, \bar{a}_j^{\mathrm{pub}}, \bar{a}_j^{\mathrm{int}}, \bar{a}_j^{\mathrm{inv}})$, intermediate inputs $\underline{a}_j$, natural

[13] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 77–80, 88.
[14] Hahnel, Democratic Economic Planning, Ch. 7, pp. 139–148, Appendix pp. 159–162, and Ch. 14, pp. 261–267.
[15] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 85–87.

resources r_j, net labor additions/releases Δl_j, net capital additions/releases Δk_j, and pollutant emissions e_j from its feasible technology set $\mathcal{T}_j$. Defining Social Benefit (SB_j) and Social Cost (SC_j):

$$SB_j(P) = p^{\text{priv}} \cdot \bar{a}_j^{\text{priv}} + p^{\text{pub}} \cdot \bar{a}_j^{\text{pub}} + p^{\text{int}} \cdot \bar{a}_j^{\text{int}} + p^{\text{cap,new}} \cdot \bar{a}_j^{\text{inv}} \tag{4.11}$$

$$SC_j(P) = p^{\text{int}} \cdot \underline{a}_j + p^{\text{res}} \cdot r_j + p^{\text{lab}} \cdot (l_j^0 + \Delta l_j) + p^{\text{cap}} \cdot (k_j^0 + \Delta k_j) + p^{\text{pol}} \cdot e_j \tag{4.12}$$

WC_j maximizes its workers' fulfillment and development $W_j(U_j, \Delta C_j)$ subject to the Socially Non-Abusive Production Constraint (PEPE Ch. 5, pp. 87–88):

$$\max_{\pi_j \in \mathcal{T}_j} \; W_j\Big(U_j(\pi_j),\, \Delta C_j(\pi_j)\Big) \quad \text{subject to} \quad SB_j(P) - SC_j(P) \geq 0 \quad \left(\text{i.e., } \frac{SB_j(P)}{SC_j(P)} \geq 1\right) \tag{4.13}$$

4.4 Critical Algorithmic Audit of pequod-cljs and Convergence Rules

In Chapter 9 of Democratic Economic Planning (2021, pp. 173–194) and Szczepanczyk (2023),[16] the authors simulated 40 experiments with $30,000$ Worker Councils, $30,000$ Consumer Councils, and 500 goods Categories (100 private, 100 public, 100 intermediate, 100 natural resources, 100 labor categories).

4.4.1 The Hahnel-Weisdorf-Szczepanczyk Price Adjustment Rule

In pequod-cljs (csvgen.clj, lines 383–463), let $S_i^{(t)}$ be total supply, $D_i^{(t)}$ be total demand, and $\text{surplus}_i^{(t)} = S_i^{(t)} - D_i^{(t)}$. Defining the normalized relative imbalance $\rho_i^{(t)}$:

$$\rho_i^{(t)} = \frac{2\left|S_i^{(t)} - D_i^{(t)}\right|}{S_i^{(t)} + D_i^{(t)}}, \qquad \tilde{\delta}_i^{(t)} = 1.05 - 0.5^{\rho_i^{(t)}} \tag{4.14}$$

the damped percentage price delta $\delta_i^{(t)}$ and price update are computed as:[17]

$$\delta_i^{(t)} = \max\left(0.001,\, \min\left(\tilde{\delta}_i^{(t)},\, \tilde{\delta}_i^{(t)} \cdot \min\left(0.25, \rho_i^{(t-1)}\right)\right)\right), \qquad P_i^{(t+1)} = P_i^{(t)}\left(1 + \text{sign}\big(D_i^{(t)} - S \tag{4.15}$$

Starting from an arbitrary uniform price vector ($P_i^{(0)} = 700$), pequod-cljs reached a 5% threshold ($\rho_i \leq 0.05\ \forall i$) in 11.85 iterations on average (19.20 iterations for

[16] Hahnel, Weisdorf, and Szczepanczyk, "Computer Simulations of Participatory Planning," Ch. 9 in Hahnel, Democratic Economic Planning, pp. 173–194; Szczepanczyk, "Pseudocode and Algorithms" (2023), pp. 43–54; source code in research/sources/pequod_cljs_csvgen_apache2.clj.

[17] Szczepanczyk, "Pseudocode and Algorithms" (2023), Eq. 9, p. 48, and pseudocode p. 50; csvgen.clj lines 383–385 and 455–463.

a 3% threshold; DEP Tables 9.1–9.2, pp. 178–179), and 6.5 iterations in Year 2 starting from Year 1's final prices.[18]

4.4.2 Four Critical Findings from Our Source-Code Audit of pequod-cljs

Direct inspection of src/clj/pequod_cljs/csvgen.clj and util.cljc reveals four crucial mathematical simplifications in the published simulations that DPEP must address:

1. Unconstrained Closed-Form Cobb-Douglas Optimization: Each WC_j maximizes $\lambda q e^c \prod_m x_m^{b_m} - \sum_m p_m x_m - s e^k$ via pre-compiled symbolic Mathematica derivatives (solution-1 through solution-8, csvgen.clj lines 209–376) without enforcing $SB_j/SC_j \geq 1$ or endowment conservation $\sum_j \Delta l_j = 0$.
2. Why "Increasing Returns to Scale" (toothaches?) Did Not Diverge: In csvgen.clj (lines 209–376), the log-linear exponent denominator is $c - k + k\sum_m b_m = k\left(\sum_m b_m + \frac{c}{k} - 1\right)$. Even when create-toothache (lines 667–672) adds +0.1 to each b_m so that physical returns to scale exceed unity ($\sum_m b_m + c > 1$), as long as the effort disutility exponent k is large enough that $\sum_m b_m + \frac{c}{k} < 1$, the council's objective function remains strictly concave! Thus, pequod-cljs did not test true non-convexities (fixed setup costs, discrete capacity jumps, or Leontief complementarities).
3. Exogenous Consumer Budgets: In consume (util.cljc lines 45–64), each Consumer Council's income (:income cc) is held constant across iterations rather than being updated dynamically from $Y^{\mathrm{GDP}}(P^{(t)})$ and worker effort hours.
4. Absence of Hard Ecological Caps: Natural resources are treated as elastically priced spot inputs against a fixed supply (1000 units) without hard barrier enforcement if demand exceeds ecological limits.

4.4.3 Algorithm 3: Proposed DPEP Safeguarded Primal-Dual / ADMM Protocol

To guarantee convergence under realistic Leontief complementarities, piecewise-linear technologies, coupled endogenous income, and hard biophysical caps, DPEP introduces Algorithm 3:

1. Proximal Trust-Region Regularization (Formalizing PEPE Ch. 4, p. 70): For round $t \geq 3$, each council's local conic/quadratic solver (Clarabel.rs) adds an Augmented-Lagrangian proximal term $\frac{\mu^{(t)}}{2}\|\pi_j - \pi_j^{(t-1)}\|_2^2$ and trust-region box constraints $|\pi_{j,i}^{(t)} - \pi_{j,i}^{(t-1)}| \leq \beta^{(t)}\pi_{j,i}^{(t-1)}$, making council supply/demand responses strictly Lipschitz-continuous even for linear Leontief technologies.

[18] Hahnel, Democratic Economic Planning, Ch. 9, Tables 9.1 and 9.2, pp. 178–179; Szczepanczyk, "Pseudocode and Algorithms" (2023), p. 51.

2. Damped Endogenous Income Fixed-Point Update:

$$w_{\text{base}}^{(t)} = (1-\alpha_w)w_{\text{base}}^{(t-1)} + \alpha_w \left(\frac{(1-\sigma_{\text{inv}})Y^{\text{GDP}}(P^{(t)}) - Y^{\text{allow}}}{H^{\text{tot}}} \right), \quad \alpha_w \in [0.4, 0.7] \tag{4.16}$$

3. Asymmetric Biophysical Barrier Multiplier: For any natural resource $r \in \mathcal{R}$ or pollutant $z \in \mathcal{Z}$ where aggregate demand breaches the hard ecological ceiling ($D_z^{(t)} > E_z^{\max}$), the price step $\delta_z^{(t)}$ is amplified by an exact barrier penalty $\gamma_{\text{eco}} \max(0, D_z^{(t)}/E_z^{\max} - 1)$, ensuring convergence only occurs strictly inside the planetary boundary $D_z^* \leq E_z^{\max}$.

4.5 Protocol State Machine and Canonical Protobuf Schemas

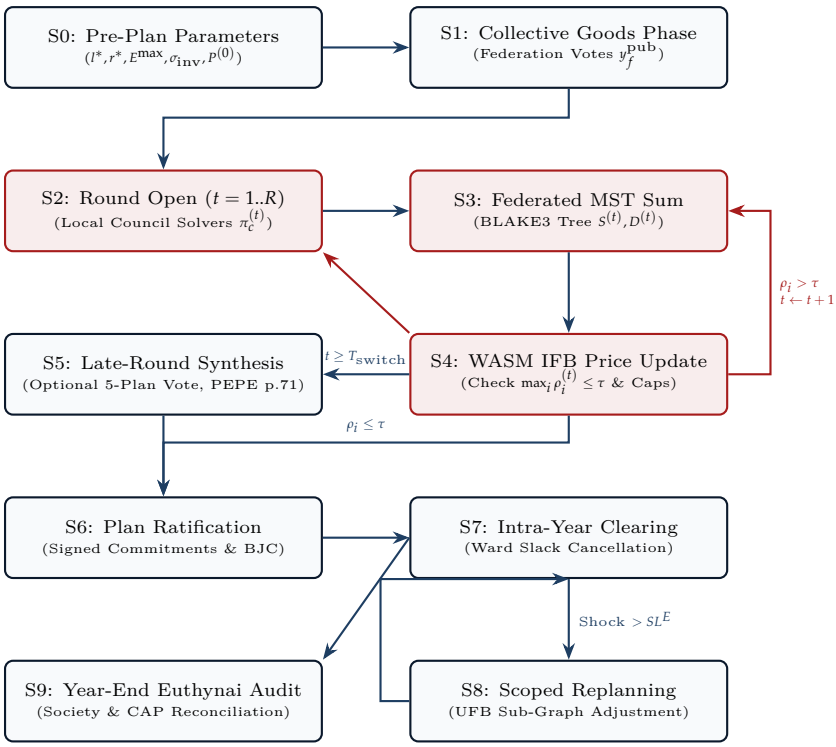


Figure 4.1: Complete 10-State DPEP Annual Planning and Intra-Year Operational State Machine

To prevent IEEE-754 floating-point non-determinism across CPU architectures (x86_64, ARMv8 smartphones, and WebAssembly), all quantities and prices in DPEP are encoded as deterministic 64-bit fixed-point integers (FixedDecimal, default scale 10^{-6} micro-units):

```
syntax = "proto3";
package dpep.v1;

message FixedDecimal {
  int64 mantissa = 1; // Value = mantissa * 10^(-scale), default scale = 6
  uint32 scale = 2;
}

message SparseItemVector {
  repeated uint32 item_ids = 1;          // Sorted catalog indices in Omega
  repeated FixedDecimal quantities = 2;  // Strictly non-negative quantities
}

message QualitativeAnnotation {          // Implements PEPE Ch. 6 'Lists' (pp. 110-112)
  uint32 item_id = 1;
  string summary_code = 2;               // e.g., "MEDICAL_NEED", "HAZARD_ABATEMENT"
  string narrative_text = 3;
  bytes content_cid = 4;                 // BLAKE3 content-addressed hash
}

message ConsumerCouncilProposal {
  uint32 plan_year = 1;
  uint32 round_number = 2;
  bytes council_id = 3;
  uint32 member_population = 4;
  SparseItemVector private_demand = 5;           // Aggregated via pairwise masking
  SparseItemVector local_public_demand = 6;
  FixedDecimal total_social_cost = 7;            // SC(CC_h, P^(t))
  FixedDecimal warranted_budget_allowance = 8;   // B(CC_h, P^(t))
  bool requests_exception = 9;                   // True iff SC > B
  repeated QualitativeAnnotation qualitative_list = 10;
  bytes member_commitments_merkle_root = 11;     // Root of private ZK budget proofs
  bytes council_threshold_signature = 12;        // Hybrid Ed25519 + ML-DSA-65 sig
}

message WorkerCouncilProposal {
  uint32 plan_year = 1;
  uint32 round_number = 2;
  bytes council_id = 3;
  uint32 industry_federation_id = 4;
  SparseItemVector proposed_outputs = 6;         // \bar{a}_j
  SparseItemVector intermediate_inputs = 7;      // \underline{a}_j
  SparseItemVector natural_resources = 8;        // r_j
  SparseItemVector total_labor_requested = 9;    // l_j^0 + \Delta l_j
  SparseItemVector total_capital_requested = 10; // k_j^0 + \Delta k_j
  SparseItemVector pollutant_emissions = 11;     // e_j
  FixedDecimal social_benefit = 13;              // SB_j(P^(t))
  FixedDecimal social_cost = 14;                 // SC_j(P^(t))
  FixedDecimal sb_sc_ratio = 15;                 // SB_j / SC_j
  bool requests_appeal = 16;                     // True iff SB_j / SC_j < 1.0
  repeated QualitativeAnnotation qualitative_list = 17;
  bytes council_threshold_signature = 18;
}
```

Listing 4.1: Canonical Protobuf Schemas for DPEP Proposals and IFB Round Checkpoints

CHAPTER 5

Distributed Systems Architecture, Consistency, and Low-Power Participation

> "Planning is computerized. All councils can access data banks of all facilitation boards and all other councils. The only thing demanding about flexible information access is that the computer system not only automatically accept but also sum and compare proposals to ascertain the status of any good."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 6, "Preparing First Proposals"[1]

5.1 Why Participatory Economics Is Not a High-TPS Blockchain Problem

When we examine the computational structure formalized in Chapter 4, three architectural truths emerge:

1. Annual Planning Is an Epoch-Synchronized Bulk-Synchronous Parallel (BSP) Process: Millions of participants and tens of thousands of councils draft proposals locally, aggregate them up nested geographic and industrial federation trees, and compute updated indicative prices $P^{(t+1)}$ once per planning round (10–20 rounds per year). Global state finality is required only at iteration boundaries and at plan ratification.
2. Intra-Year Execution Is Hierarchically Scoped: As Albert and Hahnel (1991a, Ch. 4, pp. 66–67) specify, routine intra-year consumer preference shifts cancel out at the neighborhood, ward, or city federation level within planned slack buffers ($SL^E \approx 3$–5%) without altering aggregate production plans.
3. Participation Must Work Offline and on Resource-Constrained Devices: Workers and consumers in rural collectives or intermittently connected networks must be able to deliberate, optimize local proposals, and sign ballots offline.[2]

5.2 The Four-Layer Hybrid Architecture (Architecture D)

[1] Albert and Hahnel, Looking Forward, Ch. 6 (lf6_2.htm).

[2] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan, "Local-First Software: You Own Your Data, in spite of the Cloud," Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! '19), pp. 154–178, https://doi.org/10.1145/3359591.3359737.

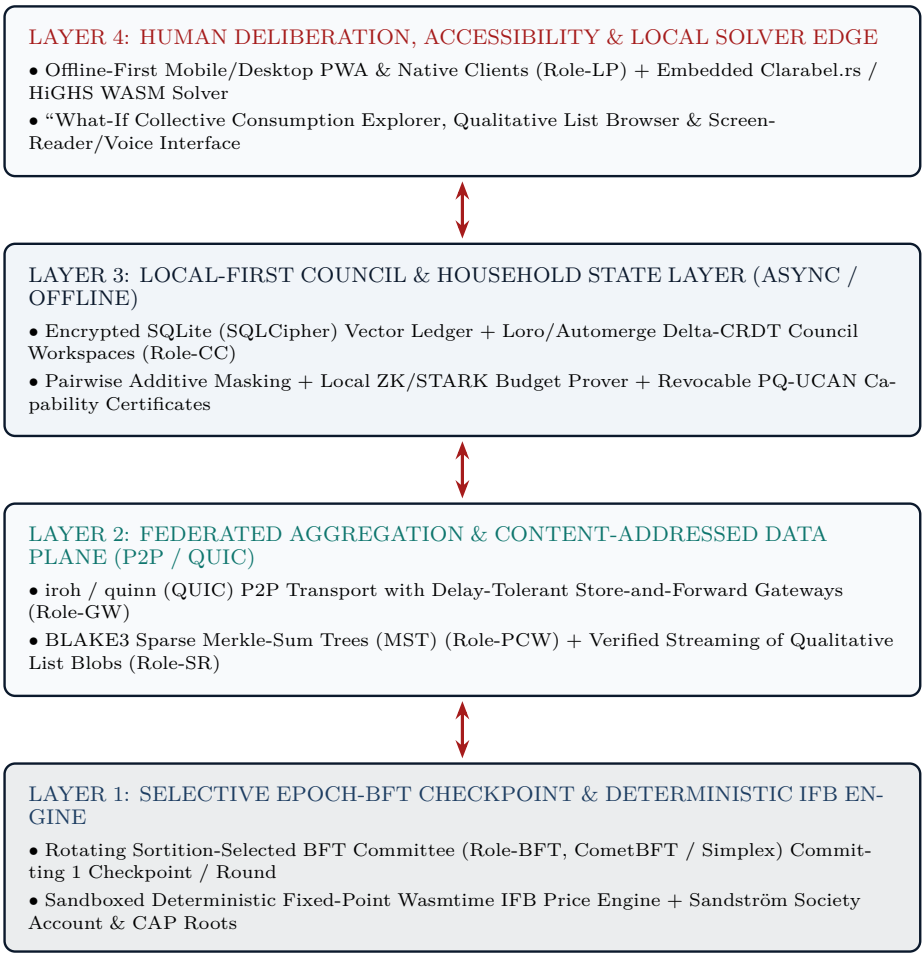


Figure 5.1: Four-Layer Architecture of the Distributed Participatory Economics Protocol (Architecture D)

5.2.1 Verifiable Federated Aggregation via Sparse Merkle-Sum Trees (MST)

In Layer 2, proposals propagate up the nested Consumer and Worker Federation trees using a BLAKE3 Sparse Merkle-Sum Tree. Every leaf node commits a council's signed proposal $\pi_c^{(t)}$ and its non-negative sparse quantity vector $v_c \in \mathbb{Z}_+^{|\Omega|}$. Every internal federation node computes:

$$\text{VectorSum}(\text{Parent}) = \text{VectorSum}(\text{Left}) + \text{VectorSum}(\text{Right}) \tag{5.1}$$

$$\text{Hash}(\text{Parent}) = \text{BLAKE3}\big(\text{Hash}_L \parallel \text{VectorSum}_L \parallel \text{Hash}_R \parallel \text{VectorSum}_R\big) \tag{5.2}$$

This construction provides a mathematical guarantee that no federation aggregator or IFB node can drop a council's proposal, alter its quantities, or inject negative phantom supply/demand: every council receives an $\mathcal{O}(\log M)$ Merkle-Sum inclusion proof against the global checkpoint root H_root.

5.3 Granular Consistency Mapping and Node Role Taxonomy

Table 5.1: Consistency Models and Partition Tradeoffs Across Economic Subsystems

Economic Subsystem	Consistency Model	Latency Target	CAP Partition Behavior	Architectural Justification
1. Proposal Drafting (S1–S2)	Strong Eventual Consistency (Delta-CRDT + SQLite)[3]	<50 ms local	AP (Offline-First): Drafts continue on local LAN/device.	Unsubmitted drafts affect no external council; offline usability is paramount.
2. Council Ratification Vote	Council-Scoped Quorum (k-of-n threshold sig)	Seconds to hours	Locally CP on Council LAN: Independent of WAN.	Only members of WC_j or CC_h hold authority over their council's proposal.
3. Federated Round Aggregation (S3)	Deterministic Convergent Tree Reduction (MST)	1–10 min per round	Bounded-Wait CP + Fallback: Carries forward $\pi_c^{(t-1)}$ if WAN severed past grace window.	Prevents a single disconnected hamlet from halting national planning while preserving verifiable inclusion.
4. IFB Price Update & Final Plan (S4, S6)	Global Total Order / Linearizable Epoch BFT[4]	2–15 s per block	CP (Safety over Liveness): Requires $\geq 2f+1$ of $3f+1$ federation validators.	Every council must optimize round $t+1$ against the exact same indicative price vector $p^{(t+1)}$.
5. Intra-Year Fulfillment (S7)	Bilateral Signed Receipts + Federated Rollup	<500 ms (offline-capable)	AP with Bounded Credit: Hardware counter prevents overspending offline.	Point-of-delivery pickups at neighborhood depots must never block on global consensus.

[3] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski, "Conflict-Free Replicated Data Types," Symposium on Self-Stabilizing Systems (SSS 2011), LNCS 6976, pp. 386–400.

[4] Ethan Buchman, Jae Kwon, and Zarko Milosevic, "The Latest Gossip on BFT Consensus," arXiv:1807.04938 (2018); Maofan Yin et al., "HotStuff: BFT Consensus with Linearity and Responsiveness," ACM PODC '19, pp. 347–356.

Table 5.2: Six-Role Hardware Realism Matrix (Supporting Legacy Smartphones and Raspberry Pis)

Node Role	Core Responsibilities	Entry Smartphone (3GB RAM ARM)?	Raspberry Pi 4/5 (4–8GB RAM)?	Community Server (8+ Cores)?
1. Lightweight Participant (Role-LP)	Local household budget QP solver, ZK budget proof, CRDT sync, MST & BFT header verifier, ballot signer.	YES (Primary) (< 50 MB RAM, $< 1\%$ battery/day)	YES	YES
2. Council Coordinator (Role-CC)	Hosts WC_j or CC_h local Wi-Fi CRDT relay, runs workplace LP/QP solver, assembles council threshold sig.	YES (Failover Hotspot)	YES (Ideal \$50 Appliance at 5W)	YES
3. Storage Replica (Role-SR)	Pins regional content-addressed BLAKE3 blobs for qualitative Lists and audit logs.	NO (Flash wear & cell data limits)	YES (with USB SSD)	YES (Primary)
4. Planning Compute Worker (Role-PCW)	Computes federated MST reductions, deterministic WASM price updates, and input-output analytics.	NO	PARTIAL (Up to $5,000$ councils)	YES (Primary IFB Node)
5. Epoch Consensus Validator (Role-BFT)	Participates in sortition-elected BFT committee finalizing round checkpoints.	NO (Requires 24/7 fixed routing)	YES (For PoC / local federations)	YES (Primary)
6. Delay-Tolerant Gateway (Role-GW)	Bridges offline mesh/LoRa/USB sneakernet bundles to federation relays.	YES (As "Data Mule")	YES (Mesh Box)	YES

CHAPTER 6

Weighted Architecture and Implementation Stack Comparison

> "Devising and choosing from among these and other possibilities is a practical issue in implementing any actual participatory economy. Assuming the procedures chosen do not violate principles essential to participatory planning, considerations include (1) the extent to which iteration workers could bias outcomes, (2) the extent of reductions in the number of iterations required to reach a plan…"
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 68[1]

6.1 Weighted Comparison of the Four Architectural Paradigms

We compare the four candidate architectures across all 14 required criteria: Architecture A (Conventional Federated SQL/REST Services), Architecture B (Pure P2P Gossip + CRDTs without finality), Architecture C (Full Per-Transaction BFT Blockchain), and Architecture D (Hybrid Local-First + Federated Merkle-Sum Trees + Selective Epoch-BFT Checkpoints).

6.2 Language and Runtime Evaluation

As detailed in Table 6.2, we select Rust as the primary systems language for dpep-core, dpep-crypto, and dpep-node—uniquely combining zero-garbage-collection memory safety, native RustCrypto NIST FIPS 203/204/205 crates, embedded interior-point conic optimization (Clarabel.rs),[3] iroh/quinn QUIC networking, Wasmtime deterministic execution, and seamless mobile/web compilation via Mozilla UniFFI (Kotlin/Swift) and WebAssembly (TypeScript PWA).

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, p. 68.

[2] Analytical derivation from Listing 4.1 schemas (at $N = 10^6$ participants, $G = 10,000$ categories): each Layer-1 RoundAggregationAndPrices checkpoint block contains 5 FixedDecimal arrays ($5 \times 10,000 \times 8$ B = 400 KB) plus 64 hybrid Ed25519 + ML-DSA-65 validator signatures ($64 \times 3,373$ B $\approx$ 216 KB) $\approx$ 616 KB/block. Over 20 annual planning rounds + 365 daily operational rollups + 50 replanning epochs (435 blocks/year), uncompressed Layer-1 growth is 435×616 KB $\approx$ 268 MB/year ($<$ 1 GB/year), while full Layer-2 council proposal archives (Role-SR) grow by $\sim$ 78 GB/year (Table 9.2). Hosting 100 federation servers at \$30–\$60/month costs \$3,000–\$6,000/month ($\approx$ \$0.003–\$0.006/citizen/month, or \$0.036–\$0.072/citizen/year). These are unbenchmarked analytical projections to be empirically tested against database indexing and P2P retransmission overhead in Chapter 9.

[3] Paul J. Goulart and Yuwen Chen, "Clarabel: An Interior-Point Solver for Conic Programs with Quadratic Objectives," University of Oxford (2024), pure Rust implementation at https://github.com/oxfordcontrol/Clarabel.rs.

Table 6.1: Weighted 14-Criterion Comparison of Architectures A, B, C, and D (1–5 Scale)

Criterion	Wt.	Arch A (REST)	Arch B (P2P)	Arch C (Chain)	Arch D (Hybrid)	Engineering & Normative Rationale
1. Decentralisation	9%	2.0	5.0	3.0	4.5	Arch A centralizes DB admin power; Arch C centralizes heavy validators; Arch D combines local autonomy with rotating BFT.
2. Trust Assumptions	10%	2.0	3.0	4.5	4.5	Arch D matches blockchain cryptographic verifiability via MST inclusion proofs + deterministic WASM at 100× lower cost.
3. Censorship Resistance	8%	2.0	4.5	4.0	4.5	Signed submission receipts allow any censored council to prove omission directly to the Epoch BFT committee.
4. Throughput & Latency	8%	5.0	4.0	2.0	5.0	Daily operations clear locally in < 10 ms; 100k council proposals aggregate in parallel in < 5 s.
5. Network Bandwidth	7%	4.5	2.5	1.5	4.5	Tree reduction + Zstd price deltas avoid $\mathcal{O}(N^2)$ gossip/block flooding.
6. Hardware Requirements	8%	4.5	3.5	1.5	4.5	Runs on $50 Raspberry Pi (Role-CC) and legacy Android (Role-LP).
7. Offline Capability	9%	1.5	5.0	1.0	5.0	Native SQLite + Loro/Automerge CRDTs + DTN store-and-forward.
8. Byzantine Fault Tolerance	8%	1.0	2.5	5.0	4.5	MST conservation checks + Epoch BFT prevent equivocation and phantom supply/demand injection.
9. Privacy	9%	2.5	2.0	1.0	4.5	Public ledgers (Arch C) leak personal consumption; Arch D uses pairwise masking + local ZK budget proofs.
10. Post-Quantum Readiness	6%	4.0	3.5	2.0	4.5	Edge MST verification absorbs 3.3 KB ML-DSA-65 signatures without bloating global blocks.
11. Implementation Simplicity	6%	5.0	3.0	2.5	3.5	Modular 4-layer stack allows clean phased rollout.
12. Operational Cost	4%	4.0	4.5	2.0	4.5	Analytical projection: < 1 GB/yr L1 checkpoint growth (~ 78 GB/yr L2 archive); $\sim$ $0.005/citizen/mo ($\sim$ $0.05/yr) server hosting.[2]
13. Energy Efficiency	4%	4.5	4.0	2.5	4.5	Epoch-driven consensus uses $\sim 10^{-7}\times$ Proof-of-Work energy.
14. Maintainability	4%	4.0	3.5	2.0	4.0	Economic rules compiled to upgradable WASM modules.
WEIGHTED SCORE	100%	3.06	3.66	2.47	4.56	Architecture D wins decisively.

Table 6.2: Programming Language Comparison Across Core Protocol Requirements (1–5 Scale)

Technical Dimension	Rust	Go	TypeScript	Kotlin	Swift	WASM (from Rust)
Memory Safety & Zero GC Pauses	5.0	3.5	2.5	3.0	4.0	5.0 (Sandboxed linear memory)
NIST PQC (ml-kem, ml-dsa, slh-dsa)	5.0	4.0	2.0	2.5	2.5	5.0 (Direct compilation)
Local-First & P2P (iroh, CRDT, SQLite)	5.0	4.5	4.0	3.0	3.0	4.5 (Browser worker target)
Operations Research (Clarabel.rs, HiGHS)	4.5	2.5	2.0	2.5	2.5	5.0 (Runs on every phone/PC)
Cross-Platform Mobile Reuse (UniFFI)	5.0	3.0	4.5	3.0	1.5	5.0 (Universal execution)
Accessible UI & Screen-Reader Tooling	2.5	1.5	5.0	4.5	4.5	Paired with TypeScript PWA UI

CHAPTER 7

Post-Quantum Cryptography, Democratic Membership, and Privacy

> "Decisions about what individuals wish to consume will be subject to collective criticism by fellow council members, though with specific guarantees for preserving individual freedoms and privacy."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 3, "Consumption Norms"[1]

7.1 Standardized Post-Quantum Cryptography (NIST FIPS 203, 204, 205)

To protect multi-decade household consumption and health-need records against "harvest-now, decrypt-later" quantum adversaries while avoiding unsubstantiated claims of unconditional lattice security, DPEP employs cryptographic agility and hybrid classical + post-quantum constructions based on the NIST standards finalized in August 2024:[2]

7.2 Democratic Membership, Sybil Resistance, and Social Trust

Design Principle 7.1 (Non-Equivalence of Cryptographic Keys and Democratic Eligibility). A cryptographic public key does not establish that its holder is a unique human being, an eligible council member, or a legitimate recallable delegate. Furthermore, in a classless participatory economy, Sybil resistance can never rely on Proof-of-Stake token wealth or Proof-of-Work energy waste.

DPEP solves the Sybil problem through Institutional Co-Attestation coupled with Sandström's Accounting Conservation Invariant:

---

[1] Albert and Hahnel, Looking Forward, Ch. 3 (lf3_2.htm); Michael Albert, Parecon: Life After Capitalism (Verso, 2003), Ch. 12 ("Consuming") and Ch. 18 ("Privacy / Frenzy: A Busybody Economy?"), where Albert specifies that individual consumption proposals within warranted effort-rated budgets are submitted anonymously to prevent "busybody" intrusion; see also Jan Philipp Dapprich, "Democratic Theory and Economic Planning: A Critique of Participatory Economics," Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024), https://ejpe.org/journal/article/view/867.

[2] National Institute of Standards and Technology (NIST), FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM), FIPS 204: Module-Lattice-Based Digital Signature Standard (ML-DSA), and FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA) (Gaithersburg, MD: NIST, August 13, 2024), https://doi.org/10.6028/NIST.FIPS.203, https://doi.org/10.6028/NIST.FIPS.204, https://doi.org/10.6028/NIST.FIPS.205.

Table 7.1: Post-Quantum Cryptographic Primitives, Byte Sizes, and ARMv8 Mobile Latency

Primitive & Standard	Security	Pub Key	Cipher/Sig	ARM Sign	ARM Verify	Assigned Role in DPEP
X25519 + ML-KEM-768 (FIPS 203 Hybrid KEM)	NIST L3 + Classical	1,216 B	1,120 B	$\sim 80\ \mu s$	$\sim 85\ \mu s$	All QUIC transport channels & Council MLS group encryption.
Ed25519 + ML-DSA-65 (FIPS 204 Hybrid Sig)	NIST L3 + Classical	1,984 B	3,373 B	$\sim 230\ \mu s$	$\sim 175\ \mu s$	Workhorse Signature for council proposals, PQ-UCAN tokens & receipts.
SLH-DSA-SHA2-128s (FIPS 205 Hash-Based)	NIST L1 (Hash Only)	32 B	7,856 B	~ 120 ms	~ 2.5 ms	Root Anchor for Federation accreditation & WASM rule signing (lattice-independent).
BLAKE3-256 (Merkle-Sum & Content CIDs)	128-bit Grover	N/A	32 B	$<$ 1 μs/KB	$<$ 1 μs/KB	Sparse Merkle-Sum Trees & content-addressed List blobs.

1. Physical Residency and Workplace Onboarding: A citizen is onboarded via threshold attestation (k-of-n) of their local Neighborhood Consumer Council (CC_h, verified against Household Facilitation Board housing records) and their Worker Council (WC_j, verified against Employment Facilitation Board rosters).

2. Why Phantom Digital Sybils Yield Zero Purchasing Power: Suppose a corrupt Neighborhood Council CC_h fabricates 500 phantom "Sybil accounts." Under Equation (4.8) (Chapter 4), consumption allowances are not a free per-capita airdrop: able-bodied adult budgets must be backed by cryptographically signed effort-hour certificates issued by accredited Worker Councils WC_j, whose total effort pool is strictly capped by $SB_j(P)/SC_j(P)$ of real outputs delivered to other councils! Dependent/disability allowances require Ward Consumer Federation audit. Thus, inventing phantom digital keys in CC_h creates zero additional purchasing power!

7.3 Privacy-Preserving Household Consumption Aggregation

How can Neighborhood Council CC_h ($m \approx 100$ households) compute its exact aggregate private demand vector $y_h^{\text{priv}} = \sum_{u=1}^{m} y_u^{\text{priv}}$ and verify that every household obeyed its budget constraint $p^{\text{priv}} \cdot y_u^{\text{priv}} + \text{PubShare}_u \leq B_u$ without any neighbor or council coordinator seeing what household u ordered?

DPEP combines two lightweight primitives executed once per planning round

on the user's smartphone (Role-LP):

1. One-Round Pairwise Additive Masking (Bonawitz Secure Aggregation): Each household u masks its private vector y_u^{priv} with pairwise zero-sum masks derived via X25519MLKEM768 key agreement ($\sum_{u=1}^{m} m_u = 0 \pmod q$), taking **1.2** ms on ARMv8. When the neighborhood coordinator sums the masked vectors, the masks cancel out, revealing only the neighborhood sum $y_h^{\text{priv}} = \sum_{u=1}^{m} y_u^{\text{priv}}$.
2. Local Zero-Knowledge Budget Compliance Proof: Household u generates a non-interactive proof Π_u (using either classical Bulletproofs in 45 ms / 1.3 KB or post-quantum hash-based Plonky3 FRI/STARKs in 180 ms / 32 KB) proving against their homomorphic commitment $C_u = \text{Com}(y_u^{\text{priv}}; r_u)$ that:

$$\forall g \in \mathcal{G}^{\text{priv}} : \; y_{u,g}^{\text{priv}} \geq 0 \quad \wedge \quad \sum_{g \in \mathcal{G}^{\text{priv}}} p_g^{\text{priv}} y_{u,g}^{\text{priv}} + \text{PubShare}_u \leq B_u \tag{7.1}$$

CHAPTER 8

Accountable AI, Mathematical Optimisation, and Failure Modes

> "Finally, we point out to those who fear that computers could become the new dictators, the computer programs are human products, socially evaluated and improved each year… In short, neither consumers nor producers need accept computer facilitated alternatives."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 9, "Revising Data in Subsequent Iterations"[1]

8.1 Architectural Rule: Deterministic Core, Advisory Edge

AI must never become an unaccountable central planner or surrogate coordinator class. DPEP enforces a strict boundary:

- Core Clearing & Pricing Engine (100% Non-AI): All price updates, SB/SC ratios, and Society Account settlements use deterministic fixed-point arithmetic and exact Operations Research (OR) solvers (Clarabel.rs / HiGHS) with verifiable Karush-Kuhn-Tucker (KKT) optimality certificates.[2]
- Edge Advisory & Accessibility Support: Classical hierarchical time-series models (ETS/ARIMA) forecast SKU size/color distributions; local QP solvers draft initial proposals from last year's bundle (eliminating David Schweickart's "consumption list burden" objection);[3] and optional local open-weight LLMs are restricted to translation, voice/screen-reader accessibility, and summarizing qualitative Lists with mandatory links to raw council text.

8.2 Critical Examination of Economic and Institutional Failure Modes

We engage directly with the most serious peer critiques of Participatory Economics—including the Winter 2024 special issue of the Erasmus Journal for Philosophy and Economics (John O'Neill, Hannes Kuch, Jan Philipp Dapprich),[4] Seth Ackerman

[1] Albert and Hahnel, Looking Forward, Ch. 9 (lf9_2.htm).

[2] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein, "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers," Foundations and Trends in Machine Learning 3, no. 1 (2011): 1–122.

[3] David Schweickart, Against Capitalism (Cambridge: Cambridge University Press, 1993), pp. 329–334.

[4] John O'Neill, "Incommensurability, Environment and Planning: A Response to Hahnel's Reply" and "Socialism, Ecology and Economic Calculation," Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024): 29–58; Hannes Kuch, "Decentralization, Federations,

Table 8.1: Evaluation of Key Decision-Support Capabilities Across Section 10 Criteria

Capability	Best Method (Is AI Needed?)	Local?	Verifiability & Challengeability	Override & Power Concentration Safeguards
1. SKU Demand Forecasting	No LLM needed: Hierarchical ETS/ARIMA + demographics.	Yes (< 50 ms)	Published MAPE error bounds & open parameters.	Non-binding baseline; councils edit freely.
2. Workplace Input Optimisation	No AI needed: Conic QP / LP (Clarabel.rs / HiGHS).	Yes (< 10 ms in WASM)	Exact KKT dual certificates prove optimality.	Presents 3 Pareto-frontier options for worker council vote; democratizes OR tools.
3. Household Proposal Drafting	No AI needed: Deterministic diff + constrained utility QP.	Yes (100% on phone)	Line-by-line diff against prior year's bundle.	Mandatory review UI highlights top changes; zero data leaves user's device.
4. Tradeoff Explanation	Deterministic Leontief Cost-Push Attribution ($(I-A)^{-1}$).	Yes (< 50 ms)	Exact algebraic decomposition of ΔP_i.	Avoids LLM hallucination by rendering exact resource/labor/pollution cost shares.
5. Qualitative List Translation	Yes: Local Neural Machine Translation + TTS.	Yes (Role-CC)	Original source text & BLAKE3 CID always linked alongside.	Bilingual delegates can post verified overrides; expands cross-regional inclusion.
6. Late-Round 5-Plan Synthesis	No AI needed: Parametric LP over 5 fixed weight vectors.	Yes (< 2 s)	Deterministic WASM LP verifiable by any council.	Councils can vote to reject all 5 plans and continue iterating (LF Ch. 9 lf9_2.htm).

(2014),[5] Paul Cockshott and Allin Cottrell (1993),[6] and David Schweickart (1993):

1. Ecological Value Incommensurability vs. PDRM Pricing ([ECON-MODEL] — O'Neill 2024 Critique): Drawing on Otto Neurath and K. William Kapp, O'Neill (2024, pp. 30–45) argues that pricing environmental damage via CAP willingness-to-accept compensation treats planetary boundaries as commensurable with consumer commodities. DPEP Resolution: DPEP enforces a strict lexicographic hierarchy: biophysical resource caps (r^*) and cumulative emissions limits ($E_z^{\max}$) set by scientific/intergenerational Environmental Planning (DEP Ch. 14) act as inviolable hard barrier constraints in Algorithm 3; PDRM compensation pricing is restricted strictly to localized, reversible nuisances below those hard ceilings.

2. Federation Centralization & Gatekeeping Against Innovation ([INST-DESIGN] — Kuch 2024 Critique): Kuch warns that Industry Federations hold gatekeeping power over accrediting new worker councils and adjudicating $SB/SC < 1$ appeals. DPEP Resolution: Any startup council denied accreditation by an incumbent Industry Federation has an automatic right of appeal to a Cross-Industry Innovation Jury (selected by VRF sortition and co-ratified by the Consumer Federation) backed by a dedicated R&D resource slice (PEPE Ch. 3, p. 42).

3. Workplace Effort Surveillance vs. Cooperative Trust ([INST-DESIGN] — O'Neill 2024 & Schweickart 1993 Critique): Critics warn that peer effort grading risks workplace humiliation or "handicap racing" gaming. DPEP Resolution: Following PEPE Ch. 3 (p. 52), DPEP supports institutional pluralism inside workplaces: while capping each council's average effort multiplier at $100 \times (SB_j/SC_j)$ externally, each WC_j may democratically choose flat equal hourly ratings ($\epsilon_u \equiv 1.0$) for all members performing their BJC hours in good faith!

4. Deliberative Fatigue and the "Too Many Meetings" Critique ([INST-DESIGN] + [SW-ARCH]): In Parecon: Life After Capitalism (2003, Ch. 18),[7] Albert addresses Nancy Folbre's concern over a "dictatorship of the sociable" and "too many meetings" by citing Pat Devine's estimate that up to 50% of advanced

and Authority in a Participatory Economy," EJPE 17, no. 2 (2024); Jan Philipp Dapprich, "Democratic Theory and Economic Planning: A Critique of Participatory Economics," EJPE 17, no. 2 (2024), full issue at https://ejpe.org/journal/issue/download/35/67.

[5] Seth Ackerman, "Horizontalism and Idealism in Socialist Imagination: An Appraisal of the Participatory Economy," Science & Society 78, no. 2 (2014): 207–239.

[6] W. Paul Cockshott and Allin Cottrell, Towards a New Socialism (Nottingham: Spokesman Books, 1993).

[7] Michael Albert, Parecon: Life After Capitalism (Verso, 2003), Ch. 18 ("Privacy / Frenzy: Dictatorship of the Sociable & Too Many Meetings?"), citing Nancy Folbre, "A Roundtable Discussion on Participatory Economics," Z Magazine (July/August 1991), Pat Devine, Democracy and Economic Planning (1988), and David Levy, review of Looking Forward, Dollars & Sense (November 1991).

capitalist GNP is consumed by adversarial transaction, surveillance, marketing, and conflict-management costs, alongside David Levy's observation that large corporations already operate massive internal non-market planning bureaucracies. To ensure participatory planning takes less time than filing taxes and paying bills under capitalism, DPEP implements a 3-Milestone Hybrid Schedule: humans deliberate at Round 1 (Priorities & Collective Goods), Round 4 (Mid-Course Review—such as a worker querying regional vs. product-failure causes of excess steel demand, as in Albert's "Sally at John Henry Steel" case study, PLAC Ch. 11), and Round 8 (Final Ratification), while local council OR solvers execute intermediate input-clearing sub-iterations automatically within council-set trust regions.

5. Dynamic Innovation & the Patent Dilemma ([ECON-MODEL] — Albert 2003, Ch. 17): Critics often claim that without millionaire patent rewards, technological innovation will stagnate. As Albert (2003, Ch. 17, "Meritocracy / Innovation") demonstrates, capitalism suffers a fundamental contradiction between stimulus to innovate (which requires patent monopolies and trade secrecy that restrict use) and static allocative efficiency (which requires immediate universal diffusion of superior techniques across all workplaces).[8] In DPEP, worker councils have a direct incentive to innovate because raising their SB_j / SC_j ratio immediately eases their work effort during the current planning cycle, while open publication of technical recipes on Layer 2 diffuses innovations across the entire Industry Federation with zero patent deadweight loss.

6. Vulnerability to Black-Market Re-Emergence ([ECON-MODEL] + [SW-ARCH] — Albert 2003, Ch. 20): Could skilled individuals trade services on the sly and accumulate capital to hire wage laborers? As Albert (2003, Ch. 20, "Market Allocation By Another Name?") shows,[9] DPEP's accounting architecture blocks black-market snowballs at three levels: (a) consumption allowances are non-transferable accounting credits rather than bearer cash, reducing illicit trade to clumsy bilateral barter; (b) any ostentatious consumption above one's warranted effort rating is immediately visible in council budget proofs; and (c) an unauthorized private enterprise cannot acquire intermediate inputs $(\underline{a}_j, r_j, k_j)$ or distribute outputs through the federated Merkle-Sum planning tree.

[8] Albert, Parecon: Life After Capitalism, Ch. 17 ("Meritocracy / Innovation"), pp. 237–244.
[9] Albert, Parecon: Life After Capitalism, Ch. 20 ("Participatory?: Market Allocation By Another Name?"), pp. 260–266.

CHAPTER 9

Proof-of-Concept Specification and Multi-Scale Benchmark Programme

> "The easiest 'experiments' to implement are a variety of computer simulations in which various features of the system and assumptions about actors' behaviors can be altered to see how outcomes change… Conceivably one could run a 'shadow' participatory economy within an existing economy."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Feasibility," pp. 108, 129[1]

9.1 Proof-of-Concept Scenario and 12 End-to-End Capabilities

The initial Proof of Concept (dpep-poc) models a self-contained, non-trivial economy of 1,000 simulated participants (plus 4 interactive human/device seats), 20 Worker Councils ($WC_1 \dots WC_{20}$ across 4 Industry Federations), 20 Neighborhood Consumer Councils ($CC_1 \dots CC_{20}$ across 4 Ward Federations and 1 City Federation), 2 Communities of Affected Parties (CAP_1, CAP_2), and 80 priced categories across all 7 classes (30 private goods, 10 public goods, 15 intermediate inputs, 8 natural resources, 7 labor categories, 5 capital asset sets, and 5 pollutants) with a 4.0% planned slack buffer (SL^E).

The PoC demonstrates all 12 required lifecycle stages end-to-end: (1) hybrid PQC council onboarding and BJC initialization; (2) collective-before-private consumption proposals with pairwise masking and ZK budget proofs; (3) constrained conic/LP worker council production proposals; (4) distributed QUIC aggregation over BLAKE3 Sparse Merkle-Sum Trees; (5) deterministic fixed-point Wasmtime IFB price updates; (6) iterative convergence to $\tau \leq 3\%$ with proximal trust regions; (7) automated verification of ecological caps and $SB/SC \geq 1$ / $SC \leq B$ invariants (including 1 special-need exception at CC_7 and 1 retooling appeal at WC_{14}); (8) council ratification voting; (9) 3-of-4 Federation threshold-signed Epoch Commitment; (10) publication of Sandström's 4-entity balance sheets; (11) 12-month intra-year execution showing ±12% neighborhood demand shifts canceling at the Ward level inside the 4% slack buffer; and (12) a Month-6 30% intermediate-input supply shock resolved via Scoped Sub-Graph Replanning (State S8) in 2 sub-rounds.

9.2 Multi-Scale Benchmark Projections (10^2 to 10^6 Participants)

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 108, 129.

Table 9.1: Mandatory Multi-Device, Mobile, Offline, and Adversarial Acceptance Tests

Test ID	Test Procedure	Measurable Acceptance Criterion
TEST-E2E-01 (Multi-Device)	Run Coordinator + 38 councils on Workstation A and WC_1 + CC_1 on Device B (Raspberry Pi / second host) over QUIC.	Converges to bit-for-bit identical H_root and P^* across both physical devices in ≤ 12 rounds.
TEST-E2E-02 (Mobile Client)	Connect a real smartphone running the Local-First PWA / WASM client (Role-LP) as a member of CC_1.	Solves household QP, generates budget proof in < 250 ms, signs with hybrid PQ key, and uses < 50 KB/round.
TEST-E2E-03 (Offline Resync)	Disconnect CC_2 during Round 2; edit proposals & qualitative Lists offline in SQLite/Loro CRDT; reconnect before Round 3 close.	Syncs CRDT deltas automatically in < 2 s with zero data loss and verifies inclusion in Round 3 H_root.
TEST-SEC-04 (Tampered/Invalid)	Inject: (a) unaccredited key, (b) bit-flipped vector, (c) $SC_h = 1.25B_h$ without exception flag, (d) negative quantity -50.	Deterministic WASM & MST verifier rejects all 4 payloads with explicit error codes and excludes from H_root.
TEST-SEC-05 (Anti-Censorship)	Fault-inject Ward Aggregator 1 to drop CC_3's valid proposal after issuing a signed submission receipt.	CC_3 detects failed MST inclusion proof, submits receipt to Epoch nodes, and triggers re-aggregation.

Methodological Note (Analytical Projections vs. Empirical Benchmarks): Only the single-JVM pequod-cljs convergence baseline (6.5–11.85 rounds; 2.5s CPU for 60,000 councils) has been empirically executed to date. All figures in Table 9.2 are analytical capacity projections derived from the byte sizes of our proto3 schemas (5.2 KB per signed sparse council proposal; 616 KB per Layer-1 checkpoint block) and micro-benchmark instruction counts. Real-world deployment will incur $1.5\times$–$3\times$ additional overhead from SQLite/CRDT indexing, P2P retransmissions, and intra-round council revisions, which the benchmark harness in Milestone M5 is designed to measure.

Table 9.2: Analytical Scaling Projections Across Five Population Scales (10-Round Annual Plan)

Metric	100	1,000 (PoC)	10,000	100,000	1,000,000
Total Councils & Federations (M)	24	51	430	4,120	40,600
Priced Categories ($G = \|\Omega\|$)	30	80	500	2,000	10,000
Council Proposals / Round	20	40	400	4,000	40,000
Total Signed Messages / Round	150	1,150	11,200	112,000	1,120,000
Aggregate Network Traffic / Round	0.8 MB	6.5 MB	68 MB	720 MB	7.8 GB
Bandwidth / Mobile Client / Round	12 KB	18 KB	35 KB	65 KB	140 KB
Archival Storage Growth / Year	10 MB	75 MB	750 MB	7.5 GB	78 GB
Mobile Client Storage (Role-LP)	< 5 MB	< 10 MB	< 20 MB	< 35 MB	< 60 MB
Solver + MST CPU Wall-Clock / Rd	< 5 ms	< 25 ms	120 ms	0.6 s	2.4 s (50 nodes)
PQC Verify + ZK CPU Wall-Clock	15 ms	110 ms	1.1 s	1.1 s (10 nodes)	2.2 s (50 nodes)
Epoch BFT Checkpoint Latency	< 50 ms	< 100 ms	0.8 s	2.0 s	3.5 s (64 nodes)
Total Automated Time / Round	$< \mathbf{0.1}$ s	$< \mathbf{0.4}$ s	$\sim \mathbf{1.5}$ s	$\sim \mathbf{4.0}$ s	$\sim \mathbf{9.0}$ s
Est. Monthly Infra Cost	\$0–\$5	\$10–\$25	\$80–\$150	\$400–\$800	\$3,000–\$6,000

CHAPTER 10

Open Theoretical, Institutional, Economic, and Technical Questions

> "A simulation incorporates more detail than a differential economic model, but still misses the richness of living interaction. As a result, the full characteristics of participatory economics and the diverse solutions to institutional problems associated with it will ultimately only reveal themselves in actual operations."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, p. 129[1]

We conclude our study by ranking the 15 most important unresolved questions across theoretical economics, institutional sociology, and distributed systems engineering, specifying how each will be investigated through either our simulation programme ([SIM]) or human-in-the-loop cooperative pilots ([HUMAN]):

Table 10.1: Ranked Research Agenda of Unresolved Questions (Priorities 1–15)

Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P1	Convergence Under Genuine Non-Convexities, Fixed Setup Costs & Leontief Complementarities	[ECON / OR] ([SIM])	Replace unconstrained Cobb-Douglas functions in pequod-cljs with mixed-integer setup costs and Leontief recipes; benchmark Algorithm 3 (ADMM proximal trust regions).
P2	Stability of Coupled Endogenous Income Feedback Across Rounds	[ECON] ([SIM])	Couple dynamic Society Account updates $(w_{\text{base}}^{(t)}, B_u^{(t)})$ into active rounds and map stability bounds of relaxation factor $\alpha_w \in (0, 1]$.
P3	Deliberative Complexity & Participation Fatigue in Multi-Round Planning	[INST / HCI] ([HUMAN])	Test the "3-Milestone Hybrid Schedule" (human rounds at $t = 1, 4, 8$ + automated intermediate solver rounds); measure NASA-TLX load.
P4	Biophysical Incommensurability vs. PDRM Pollution Pricing (O'Neill 2024)	[ECOLOGY] ([SIM/GOV])	Verify hard barrier convergence when aggregate demand hits absolute biophysical caps $(r^*, E_z^{\max})$ above local CAP PDRM pricing.
P5	Coarse-to-Granular SKU Disaggregation & Slack Buffer Sizing	[ECON / OR] ([SIM])	Simulate 50 coarse categories mapped to 2,000 SKUs under stochastic preference drift against slack $\eta \in [1\%, 10\%]$.
P6	Workplace Effort-Rating Dynamics: Solidarity vs. Surveillance or Parity	[INST] ([HUMAN])	Pilot with worker cooperatives comparing scaled effort multipliers vs. flat hourly parity ($\epsilon_u \equiv 1.0$) under BJC rotation.

Continued on next page

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, p. 129.

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Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P7	Preventing Industry Federation Cartelization & Gatekeeping (Kuch 2024)	[INST / ECON] ([HUMAN])	Test "Dual-Path Startup Accreditation" allowing rejected innovators to appeal to a sortition Innovation Jury + Consumer Federation.
P8	Calibrating Balanced Job Complexes Across Disparate Industries	[INST / OR] ([SIM/HUM])	Model cross-workplace BJC rotation as a bipartite matching problem in EFB accounting for travel & skill constraints.
P9	Inter-Regional Investment Apportionment Across Unequal Endowments	[ECON] ([SIM])	Compare per-capita vs. Porto Alegre need-weighted regional investment matrices over 10-year horizons.
P10	Reproductive & In-Home Caring Labor Integration (DEP Ch. 10)	[FEMINIST ECON] ([HUMAN])	Convene Women's and Caregivers' Caucuses to define care-allowance rules and household BJC norms.
P11	Byzantine Resilience Under Prolonged Regional Network Partitions	[DIST-SYS] ([SIM])	Benchmark Regional Proxy Extrapolation accuracy when 10% of councils are partitioned for multiple rounds.
P12	Standardization of Compact Post-Quantum Threshold Signatures	[CRYPTO] ([SW-ARCH])	Maintain CryptoSuiteId agility to transition from multi-sig bitmaps to NIST threshold ML-DSA once standardized.
P13	Privacy Anonymity Sets in Very Small Rural Hamlets ($m < 20$)	[PRIVACY] ([SW-ARCH])	Merge pairwise-masking private consumption anonymity sets at the Ward level ($m \geq 250$) while keeping collective goods local.
P14	Endpoint Security vs. Open Modifiability on User Smartphones	[SECURITY] ([SW-ARCH])	Use local hardware keystore unlock + social anomaly verification without mandatory proprietary OS attestation lock-in.
P15	Bootstrapping a "Parallel Economy" Inside Surrounding Markets	[ADOPTION] ([HUMAN])	Implement a Dual-Price Fiat Reserve Bridge in Sandström's Society Account for federated cooperatives (PEPE Ch. 6 p. 129).

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